Housing Prices in Ames, Iowa: A Multiple Linear Regression Analysis

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Introduction

Since housing is a major financial investment, its price affects homebuyers, sellers, lenders, and policymakers. Understanding how structural and neighborhood characteristics influence sale prices helps buyers make informed decisions and improves understanding of housing affordability. In this project, we examine whether gross living area, total basement area, garage area, age of house, overall quality, and neighborhoods are associated with residential sale prices in Ames, Iowa between 2006 and 2010. We hypothesize that gross living area, total basement area, overall quality, and garage area are positively associated with sale prices. We expect age of house to have a negative relationship. We also expect neighborhoods to influence housing prices. In particular, we expect overall quality to remain a significant predictor after controlling for other housing characteristics.

Previous studies show that housing characteristics are important predictors of residential property values. Using the Ames Housing dataset, Yang (2025) found that overall quality, gross living area, and garage-related features were strongly associated with sale price. These results support our predictor selection. Aziz et al. (2021) studied an urban settlement in Gujrat Pakistan. They fitted a hedonic pricing model with thirteen predictors and reported an R^2 of 0.857. Their results showed that proximity to roads and neighborhoods positively influenced housing prices. This finding supports our inclusion of these variables. Together, these studies support our variable selection. They also support our hypothesis that overall quality will remain an important predictor after controlling for other housing characteristics.

Multiple linear regression is appropriate for this project because it examines the relationship between residential sale price and multiple predictor variables simultaneously. It also allows other variables to be held constant. Our goal is to interpret how structural and neighborhood characteristics are associated with sale prices. We do not aim to maximize predictive accuracy. Hence, we focus on interpretability instead of prediction. Multiple linear regression also accommodates both numerical and categorical predictors within a single framework. Residual analysis and diagnostic plots will be used to assess assumptions of linearity, equal variance, normality, and independence. They will also help identify potential issues like outliers or non-linear patterns.

Data Description

The dataset consists of 2,930 residential sales (from Ames, Iowa) compiled by De Cock (2011) between 2006 and 2010 from the city Assessor's Office records and obtained through the AmesHousing R package (Kuhn, 2020). Originally, De Cock (2011) compiled this dataset as an alternative end-of-semester teaching exercise. However, we will analyze it to understand how structural and locational characteristics affect house prices in the housing market.

Response Variable

Sale price per residential transaction is the outcome variable, measured in U.S. dollars. Table 1 presents a distribution of sale prices, with an IQR of $129,500 to $213,500; sale prices range from $12,789 to $755,000, and the standard deviation is approximately $79,900. Figure 1 presents a histogram, showing a long-tail to the right; the mean sale price is $180,796 while the median is $160,000 — this distribution of mean Versus median sale prices can be attributed to an extremely small segment of the market on the high-end. Because sale price is a continuous and numerical measure, it is possible to model the conditional mean via ordinary linear regression.

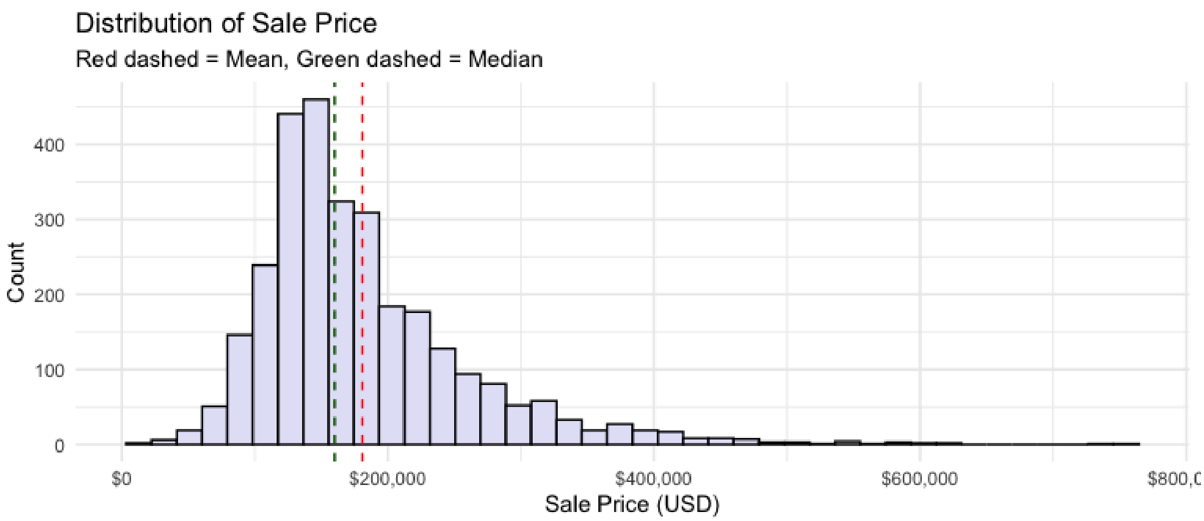


Figure 1: Histogram of Sale Price

Table 1: Statistical Summary of Sale Price

Statistics	Value
n	2930.00
mean	180796.06
sd	79886.69
median	160000.00
min	12789.00
max	755000.00
Q1	129500.00
Q3	213500.00
IQR	84000.00

Predictor Variable

Tables 2-4 and Figures 2-5 summarize the six predictors. These include Gross Living Area, Total Basement Area, Garage Area, and Age of House — all of which have a right skew with high outliers. There are homes without basements or garages that have zero values. Overall Quality has a majority from Average to Good; there are 28 Neighborhoods, but North Ames contains the majority (15%). Yang (2025) included Overall Quality, Gross Living Area, Garage Area, Basement Area as top correlates with sale price; Aziz et al. (2021) found Neighborhood & Age of House contributing to residential value. Overall, these six predictors are relevant since prior housing-price research shows that area, age, quality, and neighborhood are closely related to Sale Price.

Table 2: Definitions of Predictors and Relationships with Sale Price

Variable	Type	Definition	Relationship with Sale Price
Gross Living Area	Numerical	Above-ground living area of the house	Positive
Total Basement Area	Numerical	Total basement area of house	Positive
Garage Area	Numerical	Garage size of the house	Positive
Age of House	Numerical	Year_Sold - Year_Built, a negative value (sold before completion) was adjusted to zero.	Negative
Overall Quality	Ordinal Categorical	Overall evaluation of the house's material and finish quality, with 10 ordered categories from Very Poor to Very Excellent	Increases with quality level
Neighborhoods	Nominal Categorical	Physical locations within Ames, with 28 neighborhood levels	Differs across neighborhoods

Table 3: Statistical Summary of Numerical Predictors

Variable	mean	sd	median	min	max	Q1	Q3	IQR
Gross Living Area (sq ft)	1499.69	505.51	1442	334	5642	1126	1742.75	616.75
Total Basement Area (sq ft)	1051.26	440.97	990	0	6110	793	1301.50	508.50
Age of House (years)	36.43	30.29	34	0	136	7	54.00	47.00

Garage Area (sq ft)	472.66	215.19	480	0	1488	320	576.00	256.00

Table 4: Frequency Distribution of Categorical Predictors

Variable	Level	Count	Percent
Overall Quality	Very_Poor	4	0.14
Overall Quality	Poor	13	0.44
Overall Quality	Fair	40	1.37
Overall Quality	Below_Average	226	7.71
Overall Quality	Average	825	28.16
Overall Quality	Above_Average	732	24.98
Overall Quality	Good	602	20.55
Overall Quality	Very_Good	350	11.95
Overall Quality	Excellent	107	3.65
Overall Quality	Very_Excellent	31	1.06
Neighborhood	North_Ames	443	15.12
Neighborhood	College_Creek	267	9.11
Neighborhood	Old_Town	239	8.16
Neighborhood	Edwards	194	6.62
Neighborhood	Somerset	182	6.21
Neighborhood	Northridge_Heights	166	5.67
Neighborhood	Gilbert	165	5.63
Neighborhood	Sawyer	151	5.15
Neighborhood	Northwest_Ames	131	4.47
Neighborhood	Sawyer_West	125	4.27
Neighborhood	Mitchell	114	3.89
Neighborhood	Brookside	108	3.69
Neighborhood	Crawford	103	3.52
Neighborhood	Iowa_DOT_and_Rai	93	3.17

	l_Road		
Neighborhood	Timberland	72	2.46
Neighborhood	Northridge	71	2.42
Neighborhood	Stone_Brook	51	1.74
Neighborhood	South_and_West_of_Iowa_State_University	48	1.64
Neighborhood	Clear_Creek	44	1.50
Neighborhood	Meadow_Village	37	1.26
Neighborhood	Briardale	30	1.02
Neighborhood	Bloomington_Heights	28	0.96
Neighborhood	Veenker	24	0.82
Neighborhood	Northpark_Villa	23	0.78
Neighborhood	Blueste	10	0.34
Neighborhood	Greens	8	0.27
Neighborhood	Green_Hills	2	0.07
Neighborhood	Landmark	1	0.03

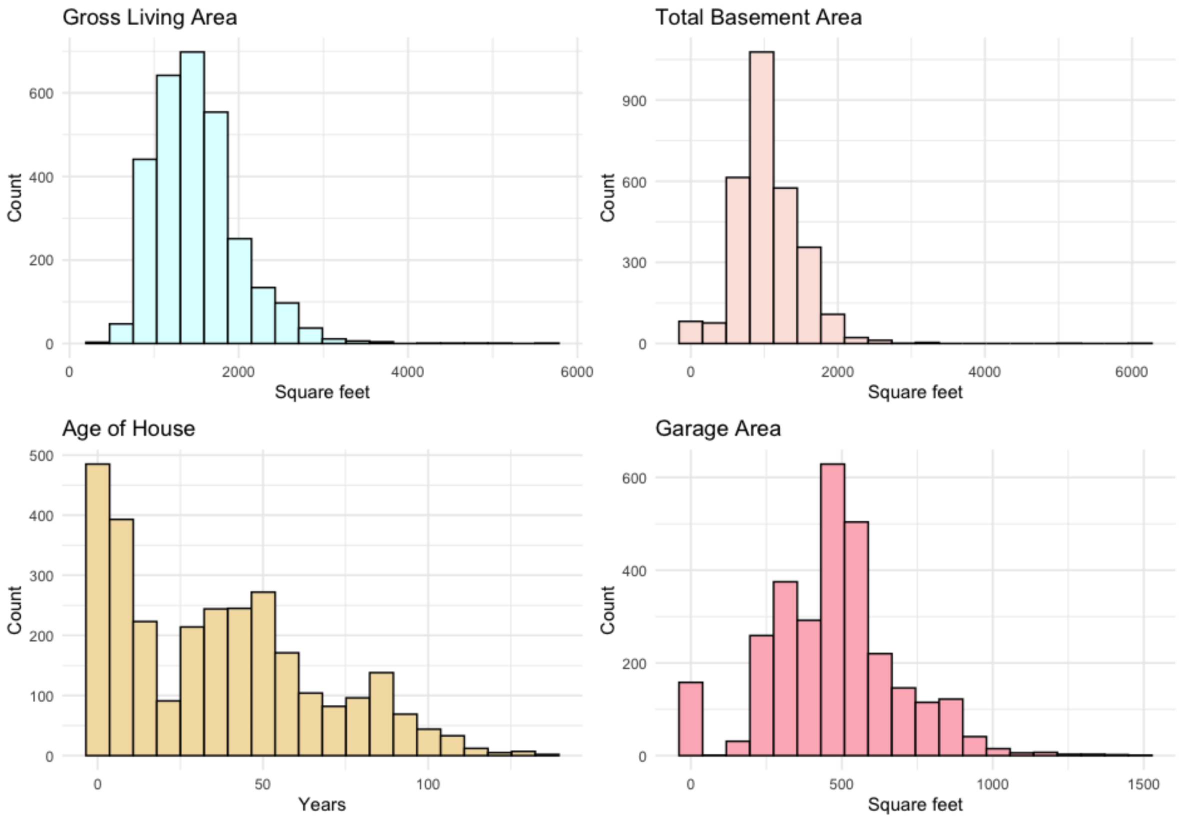


Figure 2: Histograms of Numerical Predictors

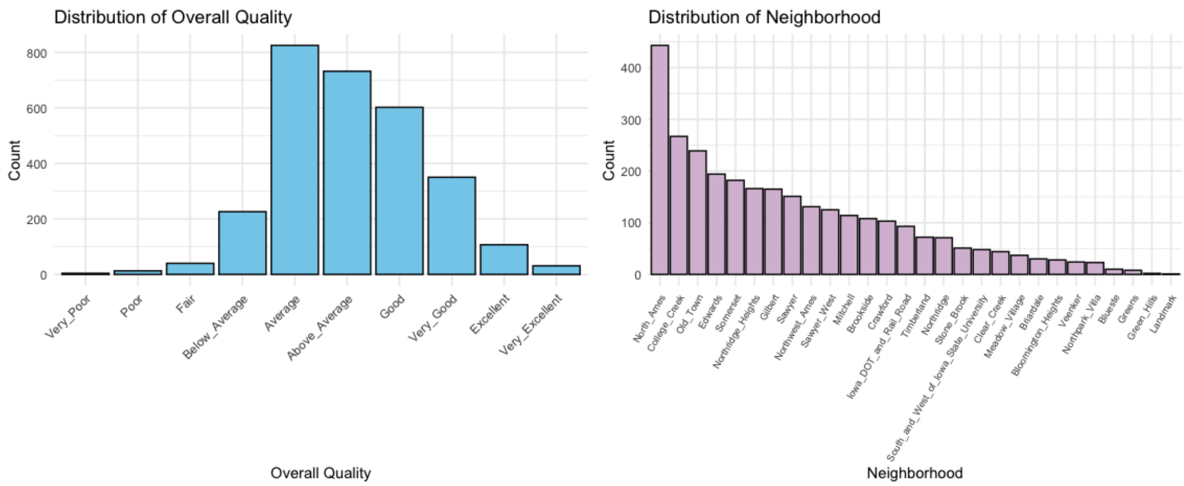


Figure 3: Bar Charts of Categorical Predictors

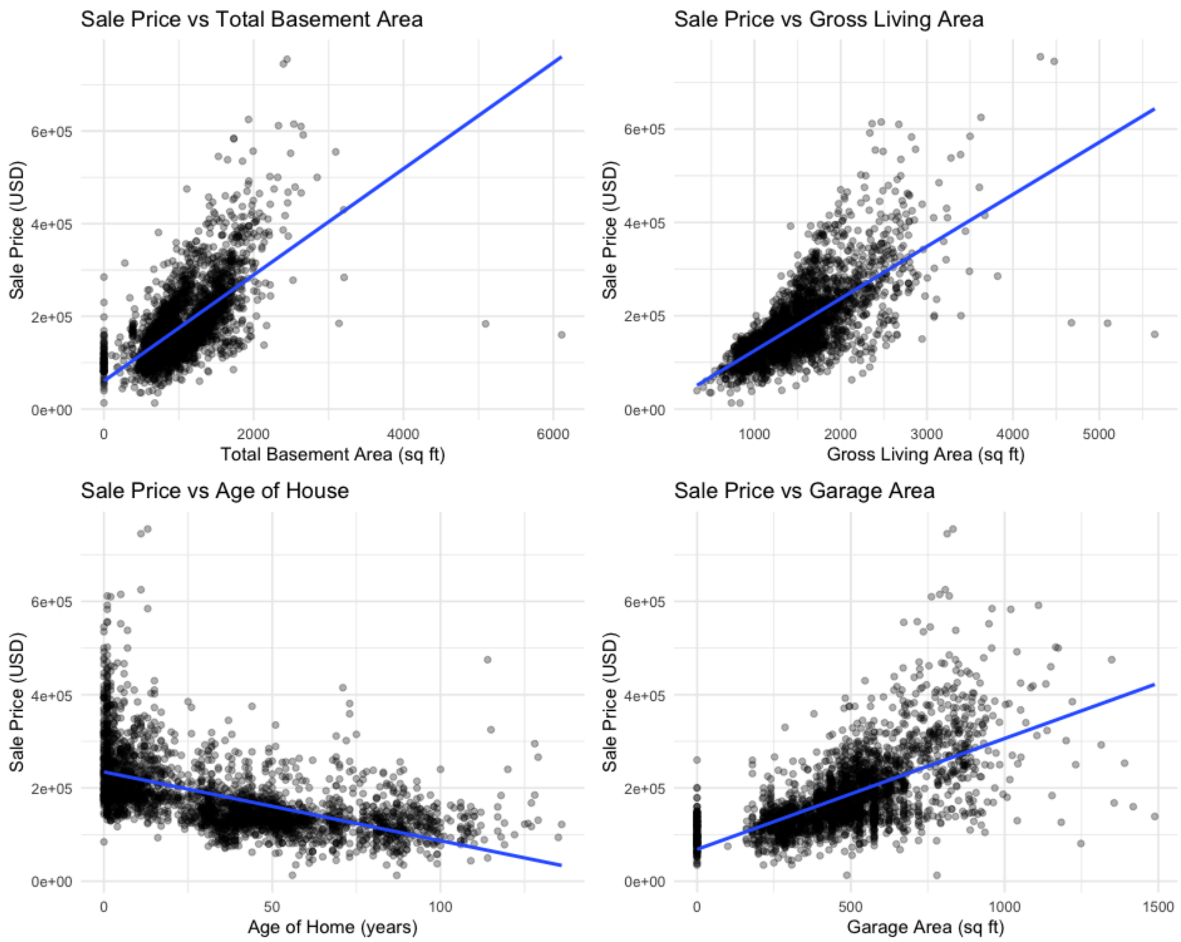


Figure 4: Scatterplots of Sale Price and Numerical Predictors

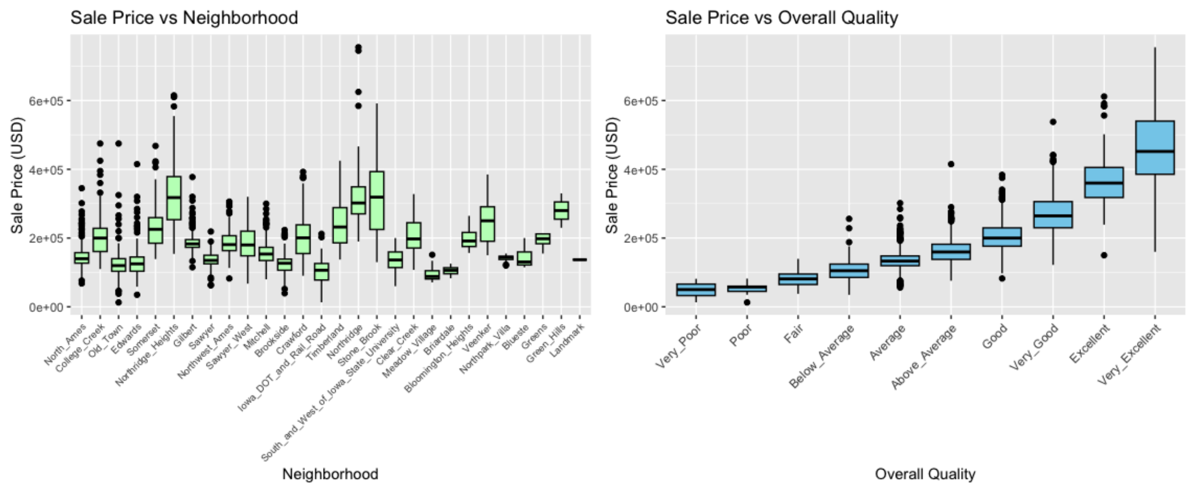


Figure 5: Boxplots of Sale Price and Categorical Predictors

Preliminary Results

Residual Analysis

We fitted a linear regression model with Sale Price as the response and 6 predictors from the AmesHousing data. The model explains 84.82% of the variation in Sale Price (Adjusted R-squared is 0.8461) and is significant overall (F-statistic = 403.4, p-value < 0.001) (Table 5).

Table 5: Preliminary Model Coefficients

(R-squared = 0.8482, Adjusted R-squared = 0.8461, F-statistic: 403.5 on 40 and 2889 DF, p-value < 0.001)

Term	Estimate	Std. Error	t value	p-value
(Intercept)	22875.35	16052.17	1.43	0.15425
Gr_Liv_Area	46.22	1.61	28.63	< 0.001
Total_Bsmt_SF	18.77	1.79	10.5	< 0.001
Overall_QualPoor	16571.93	17987.53	0.92	0.35697
Overall_QualFair	23161.71	16483.91	1.41	0.1601
Overall_QualBelow_Average	31647.36	15892.32	1.99	0.04654
Overall_QualAverage	40509.28	15816.05	2.56	0.01048
Overall_QualAbove_Average	49741.39	15850.83	3.14	0.00172
Overall_QualGood	62890.14	15934.81	3.95	< 0.001
Overall_QualVery_Good	91313.23	16106.88	5.67	< 0.001
Overall_QualExcellent	158695.27	16471.05	9.63	< 0.001
Overall_QualVery_Excellent	197274.24	17352.97	11.37	< 0.001
Age_ames	-329.55	43.23	-7.62	< 0.001
Garage_Area	35.28	3.71	9.51	< 0.001
NeighborhoodCollege_Creek	7063.32	2977.43	2.37	0.01774
NeighborhoodOld_Town	-9418.81	3013.68	-3.13	0.00179
NeighborhoodEdwards	-9029.03	2771.36	-3.26	0.00114
NeighborhoodSomerset	10012.09	3517.11	2.85	0.00445
NeighborhoodNorthridge_Heights	35918.01	3901.1	9.21	< 0.001
NeighborhoodGilbert	5525.15	3427.75	1.61	0.1071
NeighborhoodSawyer	415.94	2981.15	0.14	0.88905
NeighborhoodNorthwest_Ames	2049.08	3292.06	0.62	0.53371
NeighborhoodSawyer_West	1084.21	3491.86	0.31	0.75621
NeighborhoodMitchell	1558.47	3421.16	0.46	0.64876
NeighborhoodBrookside	2409.91	3584.88	0.67	0.50148
NeighborhoodCrawford	34088.95	3597.75	9.48	< 0.001

NeighborhoodIowa_DOT_and_Rail_Road	-10761.79	3893.34	-2.76	0.00574
NeighborhoodTimberland	22282.54	4363.58	5.11	< 0.001
NeighborhoodNorthridge	48901.43	4724.76	10.35	< 0.001
NeighborhoodStone_Brook	46119.63	5325.37	8.66	< 0.001
NeighborhoodSouth_and_West_of_Iowa_State_University	-8027.11	4993.6	-1.61	0.10806
NeighborhoodClear_Creek	27908.86	5019.71	5.56	< 0.001
NeighborhoodMeadow_Village	-19193.05	5654.52	-3.39	< 0.001
NeighborhoodBriardale	-25170.25	6051.18	-4.16	< 0.001
NeighborhoodBloomington_Heights	763.59	6511.49	0.12	0.90666
NeighborhoodVeenker	24570.28	6750.57	3.64	< 0.001
NeighborhoodNorthpark_Villa	-14470.27	6803.88	-2.13	0.03352
NeighborhoodBlueste	-17020.52	10156.1	-1.68	0.09387
NeighborhoodGreens	-3358.76	11501.35	-0.29	0.77028
NeighborhoodGreen_Hills	109730.01	22328.89	4.91	< 0.001
NeighborhoodLandmark	-21242.82	31441.98	-0.68	0.49934

Linearity is questionable: the Response Versus Fitted plot (Figure 6) deviates from the y = x line at high fitted values, and both Sale Price Versus Age (Figure 4) and Residuals Versus Age (Figure 7) show a curved pattern, suggesting the linear form is not appropriate for the Age effect. Homoscedasticity is violated: the Residuals Versus Fitted plot (Figure 6) shows a funnel shape with variance increasing for higher fitted values, and Residuals Versus Garage Area, Total Basement Area, and Gross Living Area (Figure 7) exhibit a similar pattern. Normality is questionable: the Normal (Figure 6) deviates from the reference line in both tails, with the large negative residuals in the lower tail indicating left-skewed residuals. Independence is plausible since the data consist of cross-sectional observations of distinct housing units. A log/Box-Cox transformation of Sale Price should be considered to address these violations.

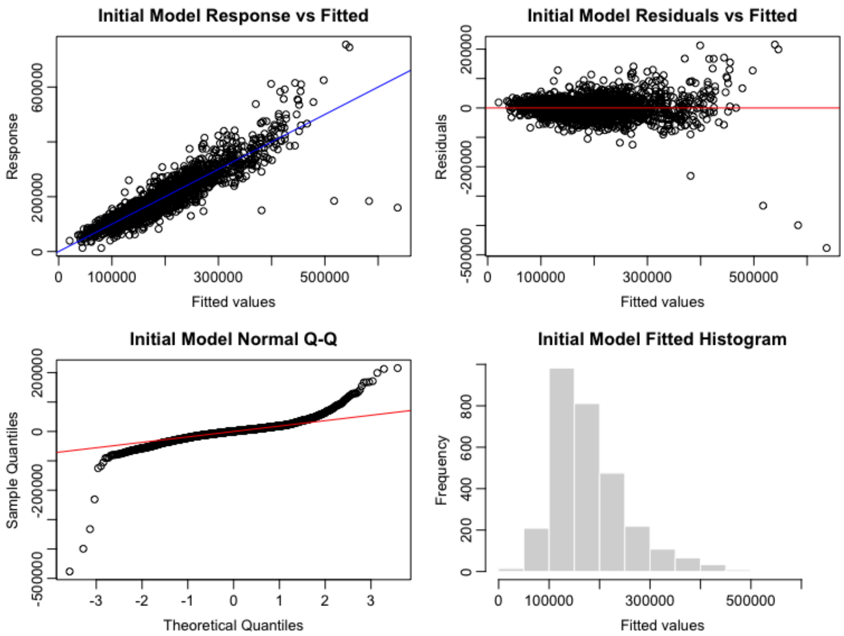


Figure 6: Response Versus Fitted Values, Residuals Plot, Fitted Histogram, Normal Q-Q Plot of Initial Model.

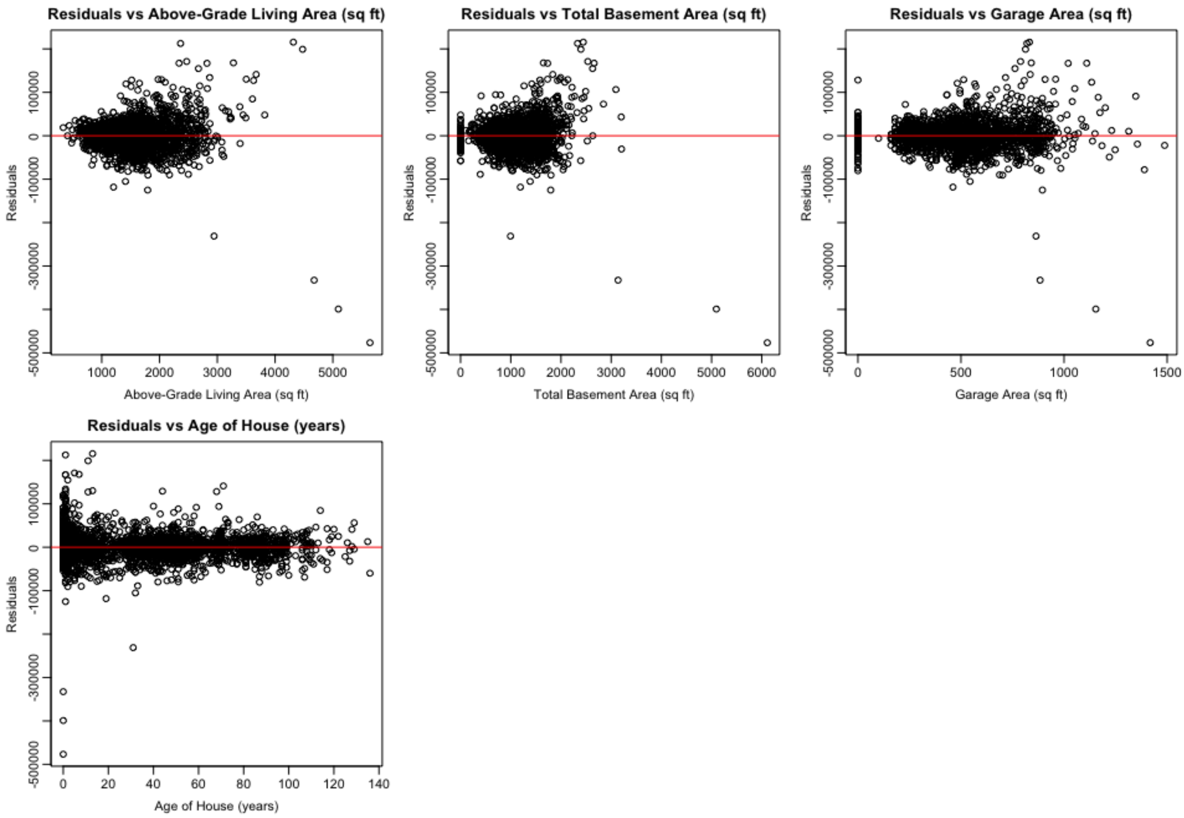
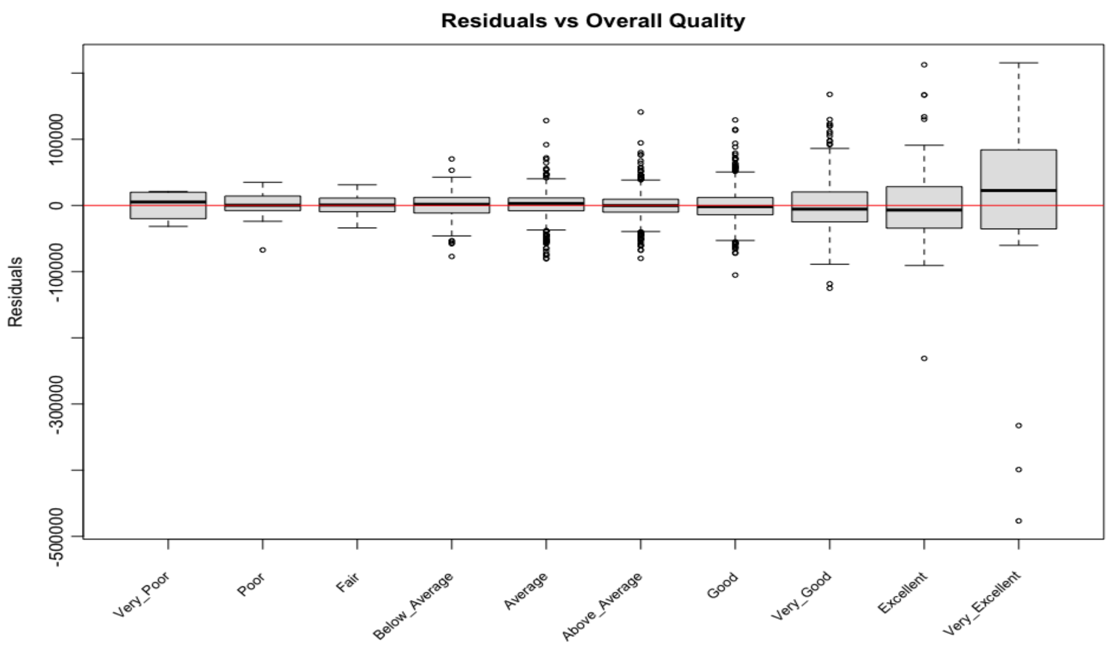


Figure 7: Residuals Versus All Numerical Predictors of the Initial Model



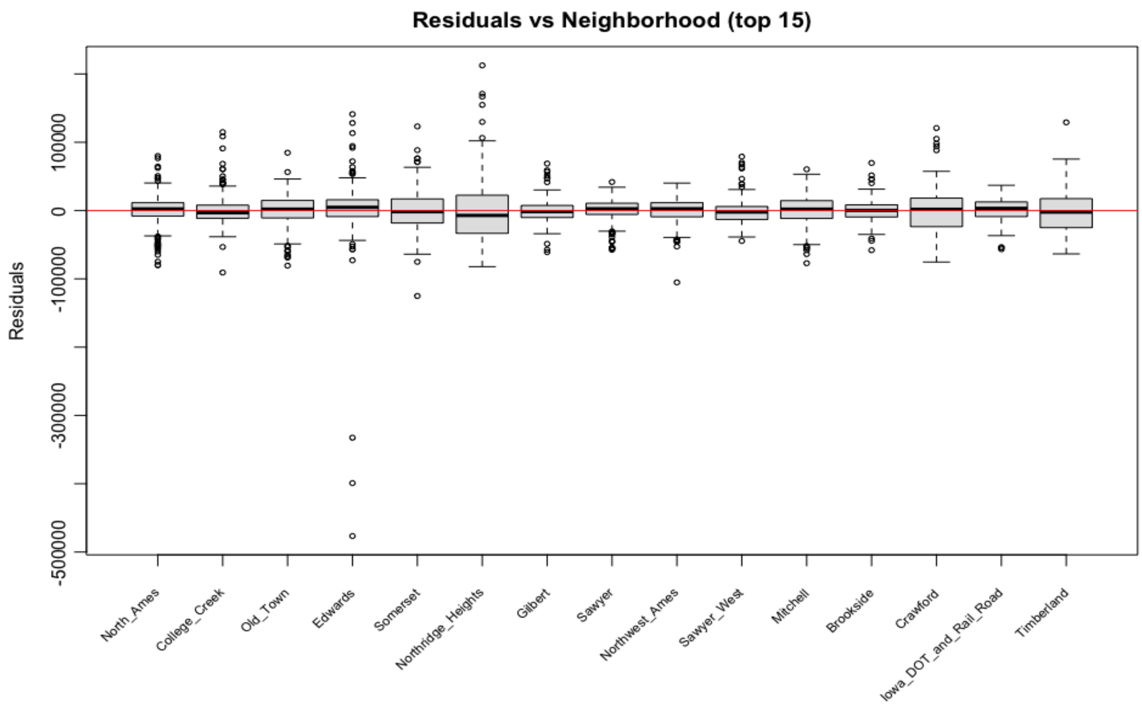


Figure 8: Residuals Versus All Categorical Predictors of the Initial Model (And Take Top 15 Neighborhoods Residual Versus Neighborhood).

Model Discussion

The coefficient estimates support the research hypothesis. Holding other predictors constant, each additional square foot of gross living area increases the estimated Sale Price by $46.22, and "Very Excellent" houses sell for an estimated $197,274 more than "Very Poor" ones. The positive estimated coefficients of Gross Living Area, Total Basement Area, and Garage Area align with Yang (2025), who concluded these as the top factors that positively correlated with house prices. The estimated coefficient for Age is negative (−$329.55 per year, p-value < 0.001), consistent with Aziz et al. (2021), who similarly attributed negative age effects to buyer preferences for newer homes. The consistency in relative importance across specifications strengthens our conclusions despite the assumption violations.

Model Selection

Initial Model: Sale $\text{Price}_i = \beta_0 + \beta_1\text{Gross Living Area}_i + \beta_2\text{Total Basement Area}_i + \beta_3\text{Garage area}_i + \beta_4\text{Age of House}_i + \beta_5\text{Overall Quality}_i + \beta_6\text{Neighborhoods}_i + \varepsilon_i$

Model 1: Sale $\text{Price}_i = \beta_0 + \beta_1\text{Gross Living Area}_i + \beta_2\text{Total Basement Area}_i + \beta_3\text{Garage area}_i + \beta_4\text{Age of House}_i + \beta_5\text{Overall Quality}_i + \beta 6\text{Neighborhood}_i + \beta_7\text{Lot Area}_i + \beta_8\text{Full Bath}_i + \beta_9\text{Bedroom Above Ground}_i + \beta_{10}\text{Fireplaces}_i + \beta_{11}\text{Total Rooms Above Ground}_i + \varepsilon_i$

Model 1 was expanded to include Lot Area, Full Baths, Bedrooms Above Ground, Fireplaces, and Total Rooms Above Ground. The preliminary model explained the general impacts of size, quality, and location; however, it did not adequately include a number of other structural elements. Chau and Chin (2003) have identified rooms, bedrooms, bathrooms, lot size, garages, and fireplaces as structural elements typically found in hedonic price models. Therefore, these structural elements were added as predictors to reflect some additional structural dimensions of housing value.

The Age of House residual plot (Figure 10) did show a very small improvement from the preliminary model. The Age of House residuals showed a slightly more centered distribution overall and the differences between the newer homes were somewhat reduced in spread from the preliminary model. All other diagnostic plots remained relatively unchanged.

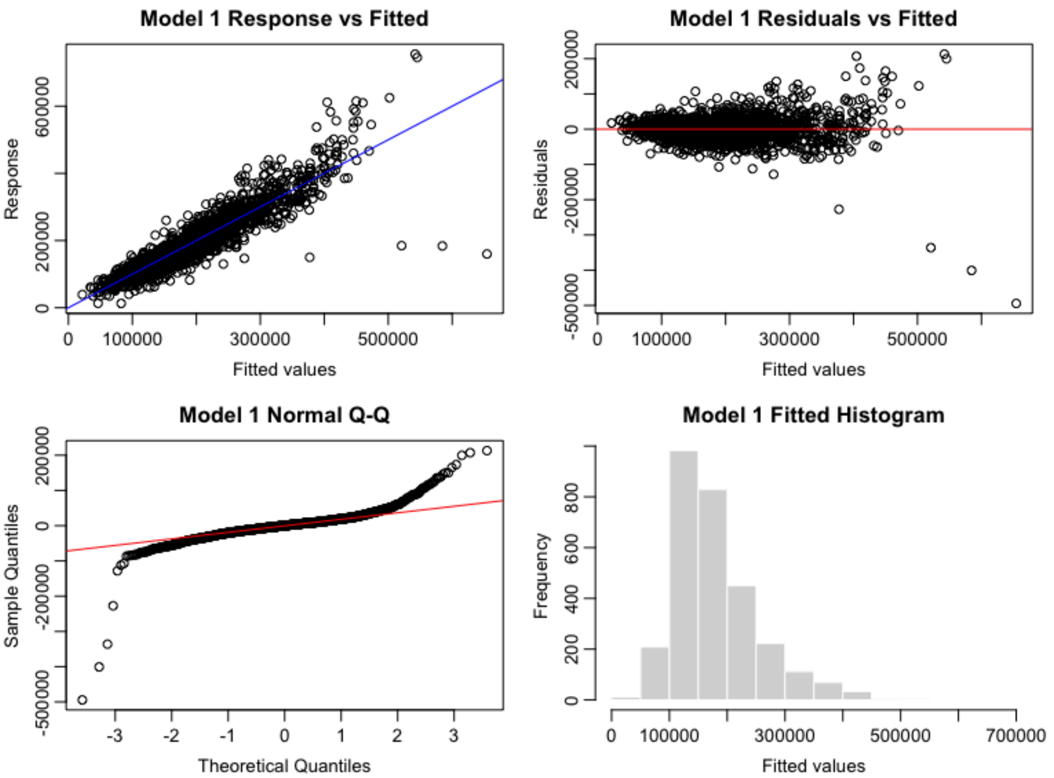


Figure 9: Response Versus Fitted Values, Residuals Plot, Fitted Histogram, Normal Q-Q Plot of Model1.

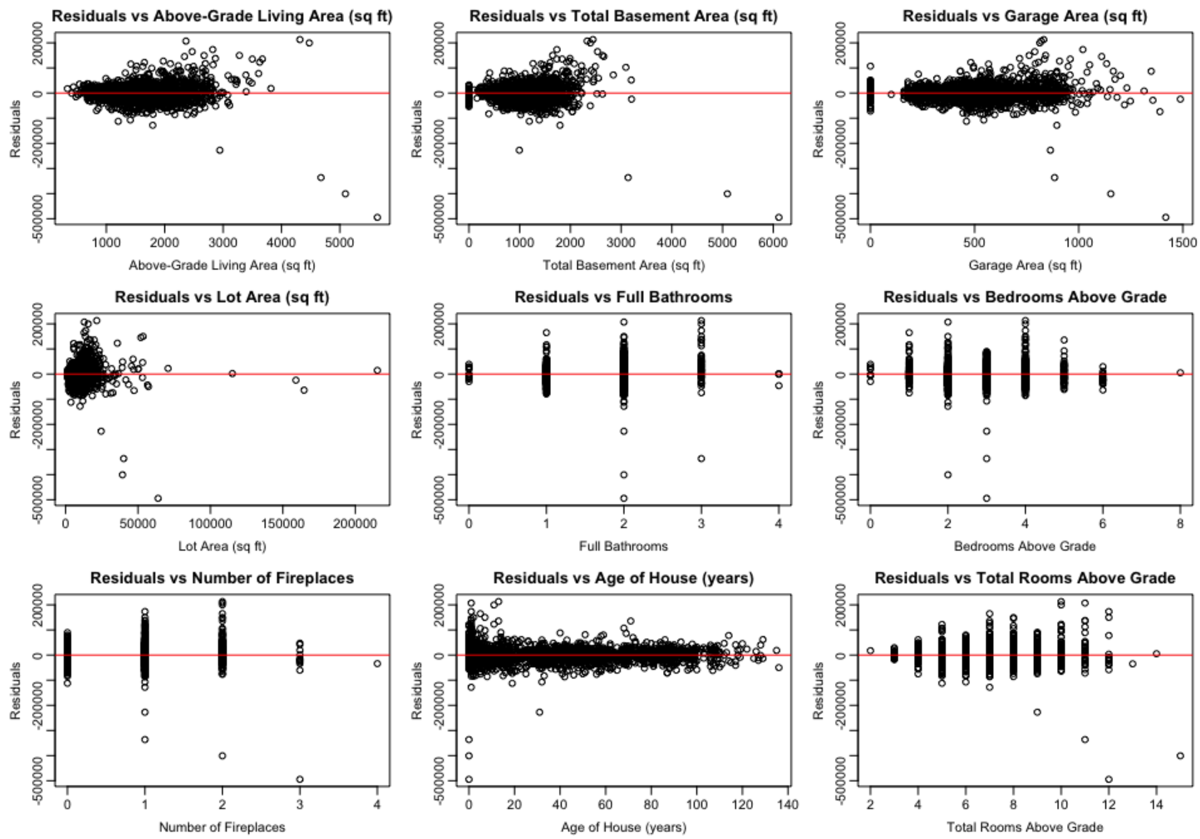
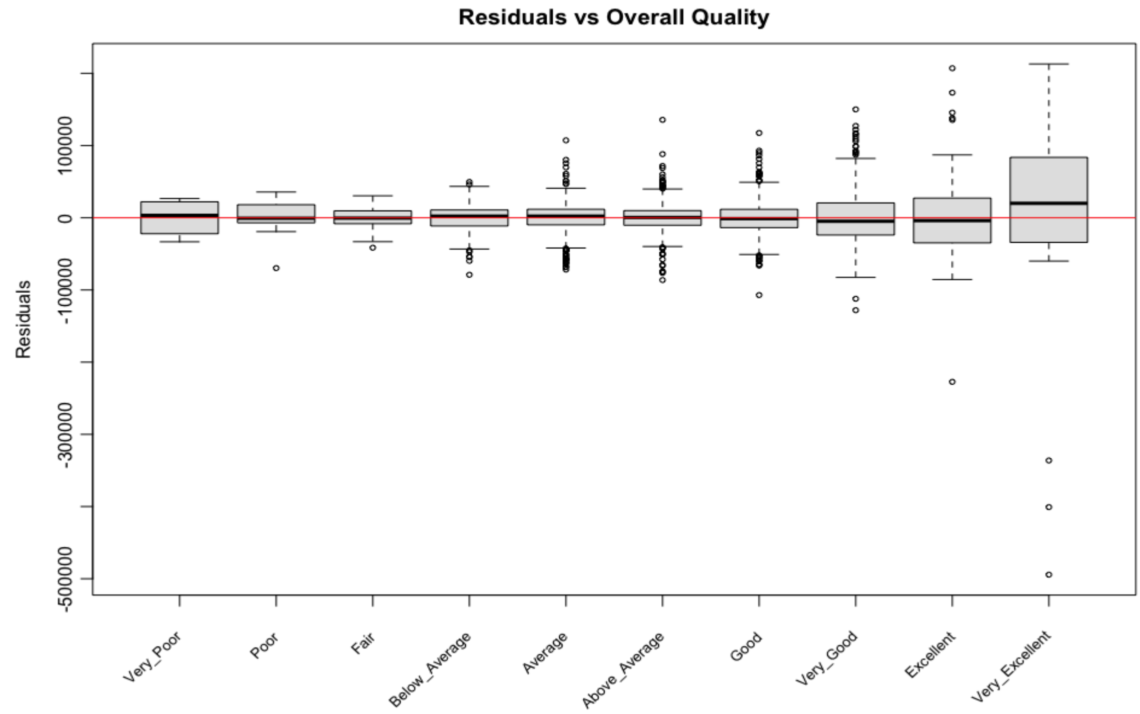


Figure 10: Residuals Versus All Numerical Predictors of Model 1.



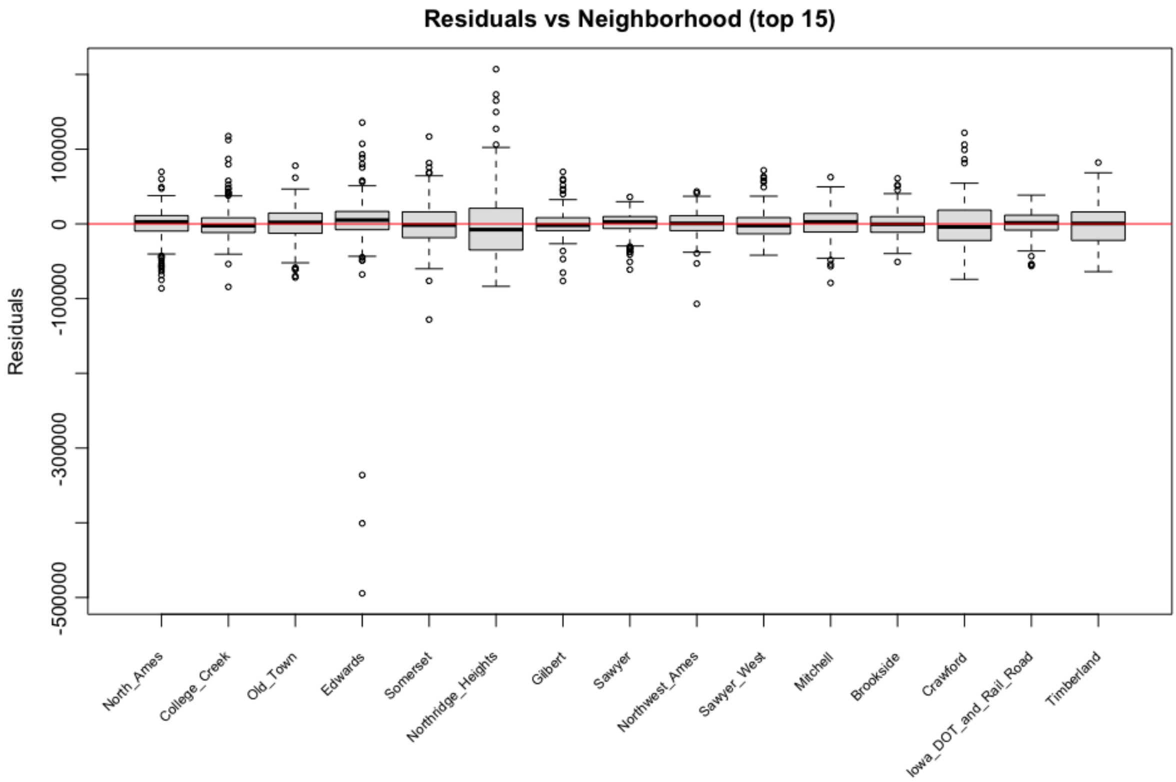


Figure 11: Residual Versus All Categorical Predictors of Model 1 (And Take Top15 Neighborhoods in the plot of Residual Versus Neighborhood).

Model 2: Box-Cox Transformation$_{\lambda=0.222}$(Sale Price$_i$)= β_0 + β_1Gross Living Area$_i$ + β_2Total Basement Area$_i$ + β_3Garage area$_i$ + β_4Age of House$_i$ + β_5Overall Quality$_i$ + β6Neighborhood$_i$ + β_7Lot Area$_i$ + β_8Full Bath$_i$ + β_9Bedroom Above Ground$_i$ + β_{10}Fireplaces$_i$ + β_{11}Total Rooms Above Ground$_i$ + ε_i

Utilizing a Box-Cox transformation (λ = 0.222) on Sale Price as the response variable for Model 2 resulted in slightly tighter relationships between the fitted and actual values as well as the residuals and the fits (see Figure 12). In the Normal Q-Q plot, the severe upper tail departures were both less pronounced than before, and the fitted histogram had changed from positively skewed to nearly symmetric. In the residual plot against Age of House (Figure 13), there appeared to be a random scattering of the residuals, therefore, the dependence of the residuals on age of the house has decreased.

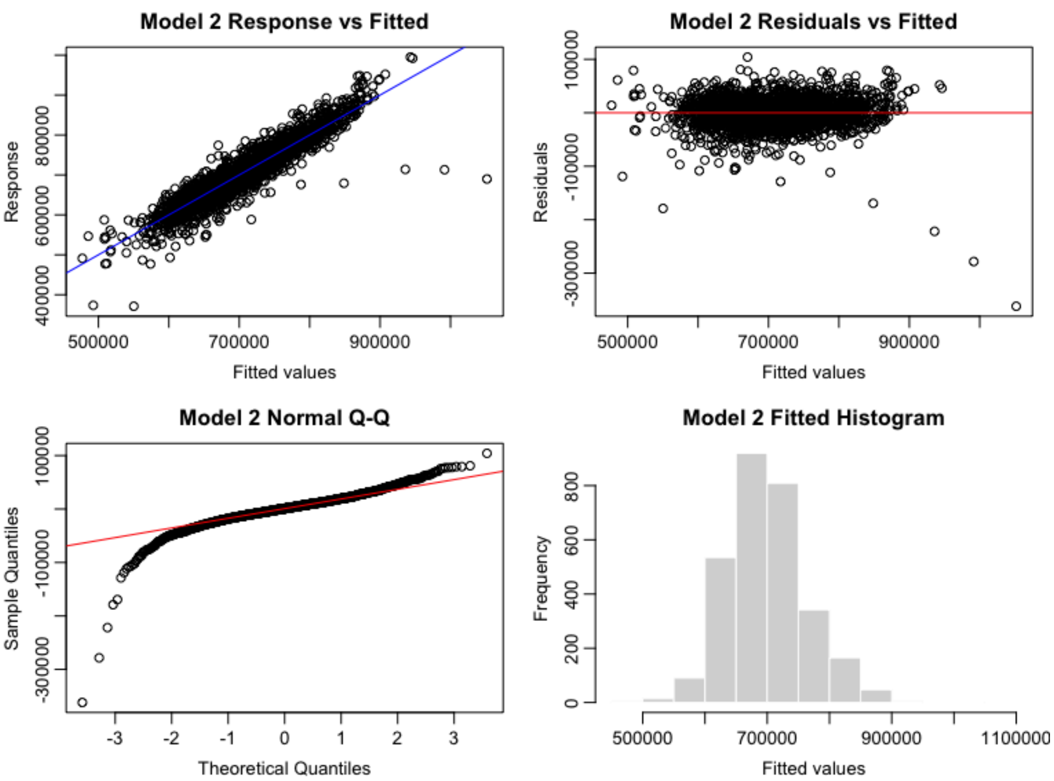


Figure 12: Response Versus Fitted Values, Residuals Plot, Fitted Histogram, Normal Q-Q Plot of Model 2.

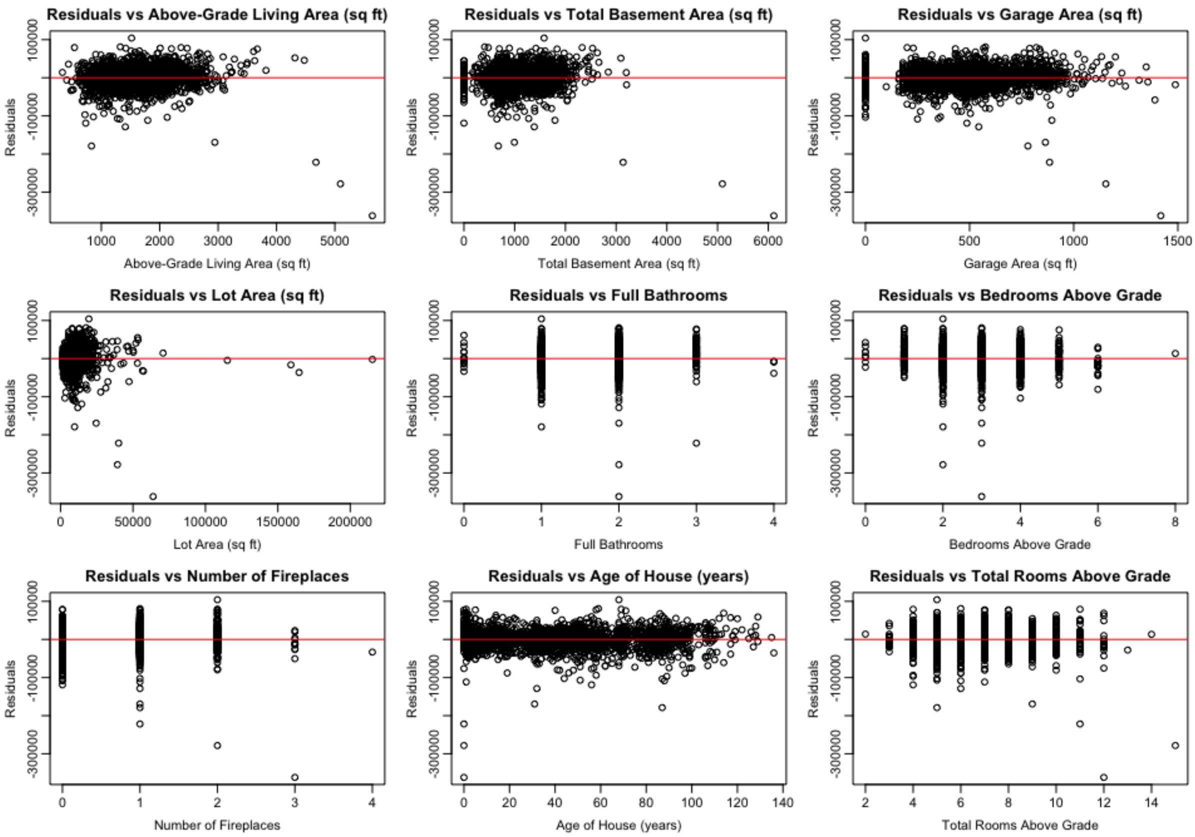
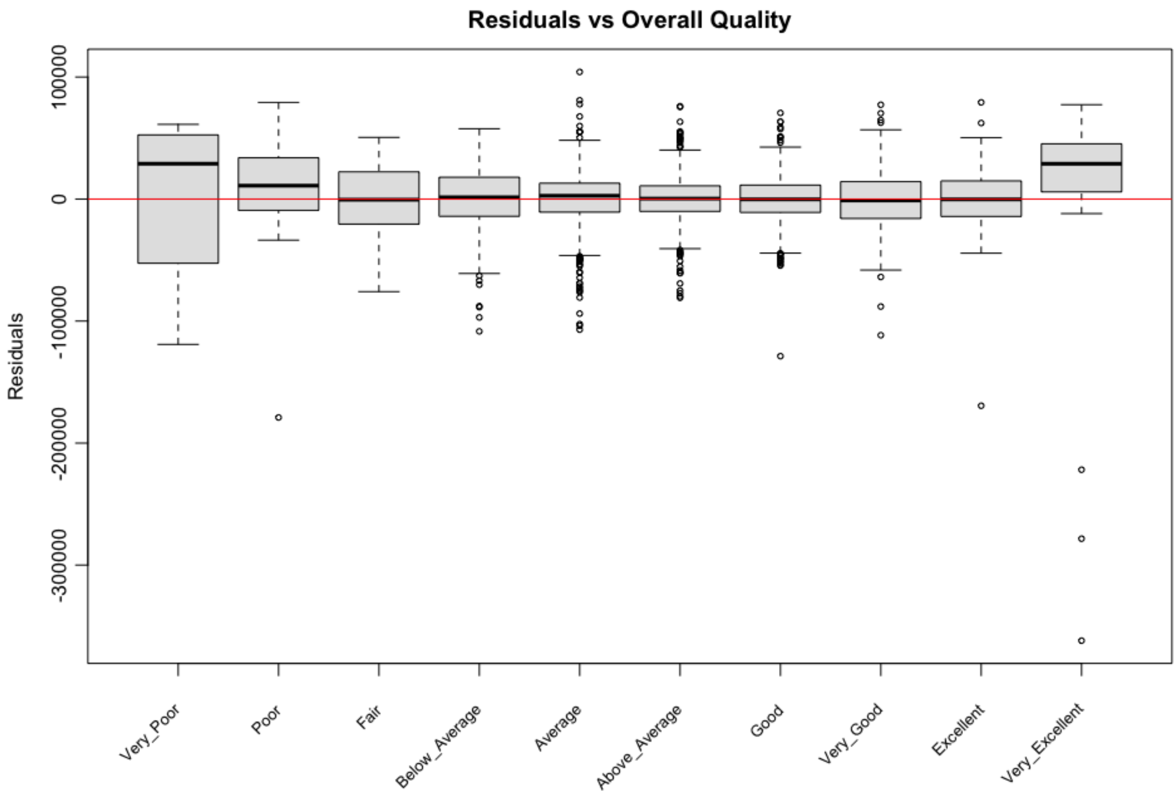


Figure 13: Residuals Versus All Numerical Predictors of Model 2.



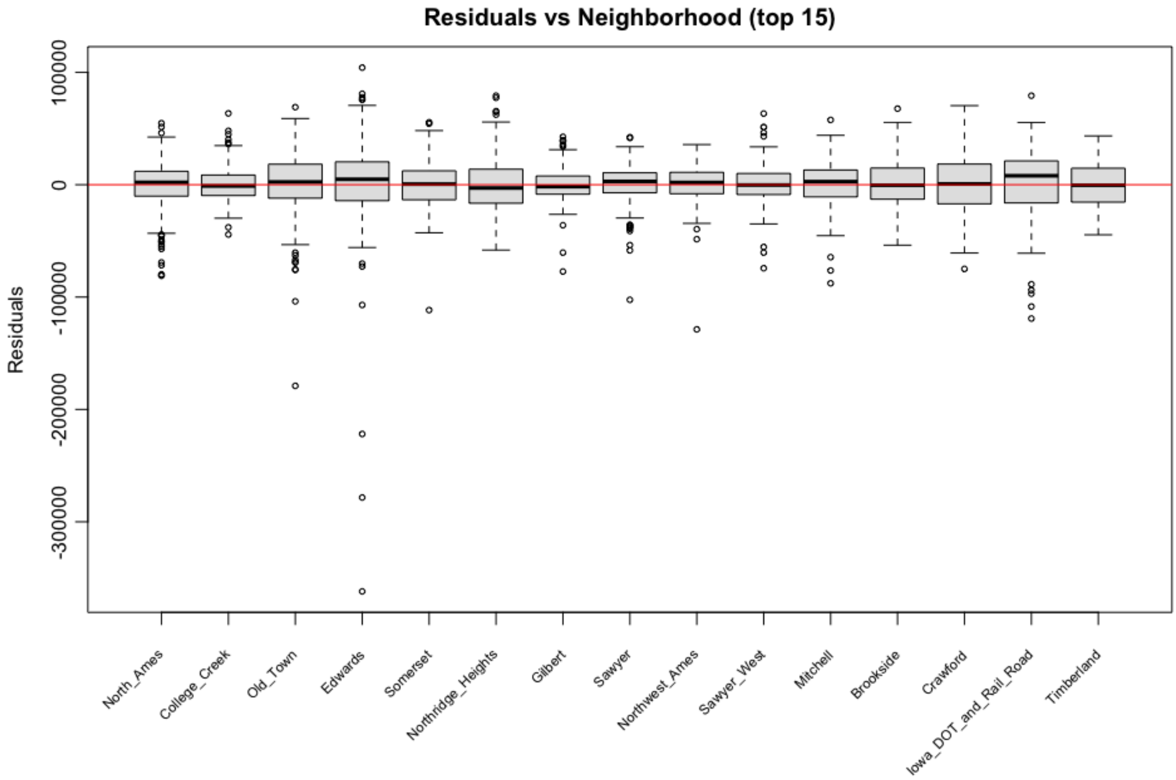


Figure 14: Residuals Versus All Categorical Predictors of Model 2 (And Take Top 15 Neighborhoods in the plot of Residual Versus Neighborhood).

Model 3: Box-Cox Transformation$_{\lambda=0.222}$(Sale Price$_i$)= β_0 + β_1Gross Living Area$_i$+ β_2Total Basement Area$_i$ + β_3Log(1 + Garage area$_i$) + β_4Age of House$_i$ + β_5Overall Quality$_i$ + β_6Neighborhood$_i$ + β_7Lot Area$_i$ + β_8Full Bath$_i$ + β_9Bedroom Above Ground$_i$ + β_{10}Fireplaces$_i$ + β_{11}Total Rooms Above Ground$_i$ + ε_i

Due to the presence of many zero values in the garage area and a considerable range of nonzero observations in the garage area dataset (Figure 13), the Log(1 + Garage Area) transformation was chosen for model 3 to provide residuals that were roughly randomly scattered about 0. Other predictors did not show strong evidence of improved fit, therefore they were not transformed.

Following model 3, a VIF check on all numeric predictors were conducted in table 6. Each of the numeric predictors had VIF values < 5, indicating that there was no evidence of severe multicollinearity.

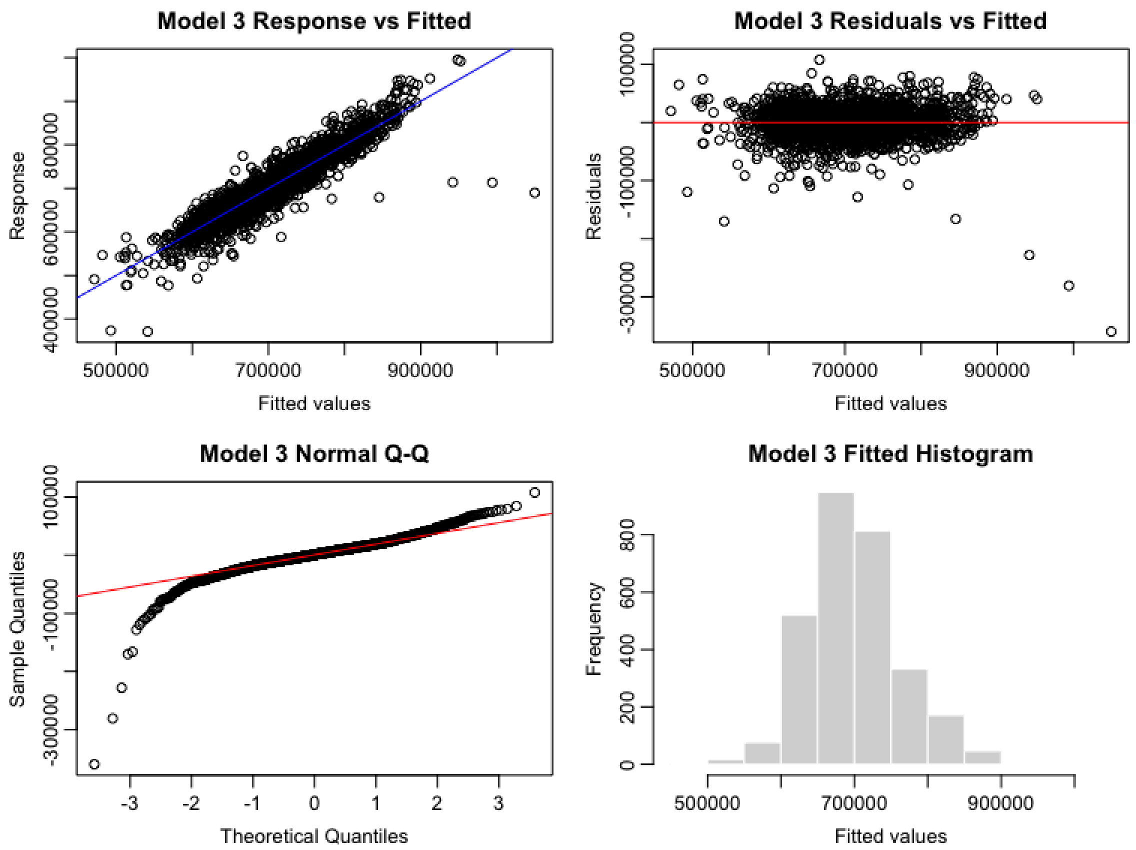


Figure 15: Response Versus Fitted Values, Residuals Plot, Fitted Histogram, Normal Q-Q Plot of Model 3.

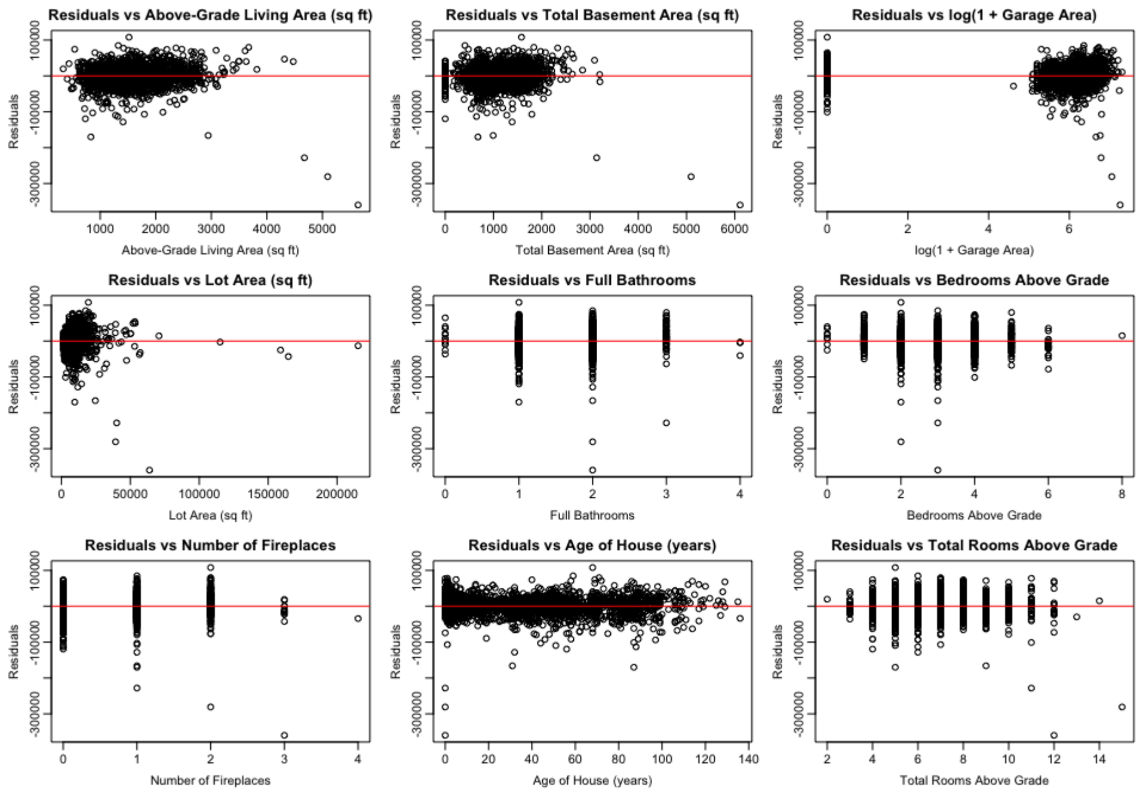


Figure 16: Residuals Versus All Numerical Predictors of Model 3.

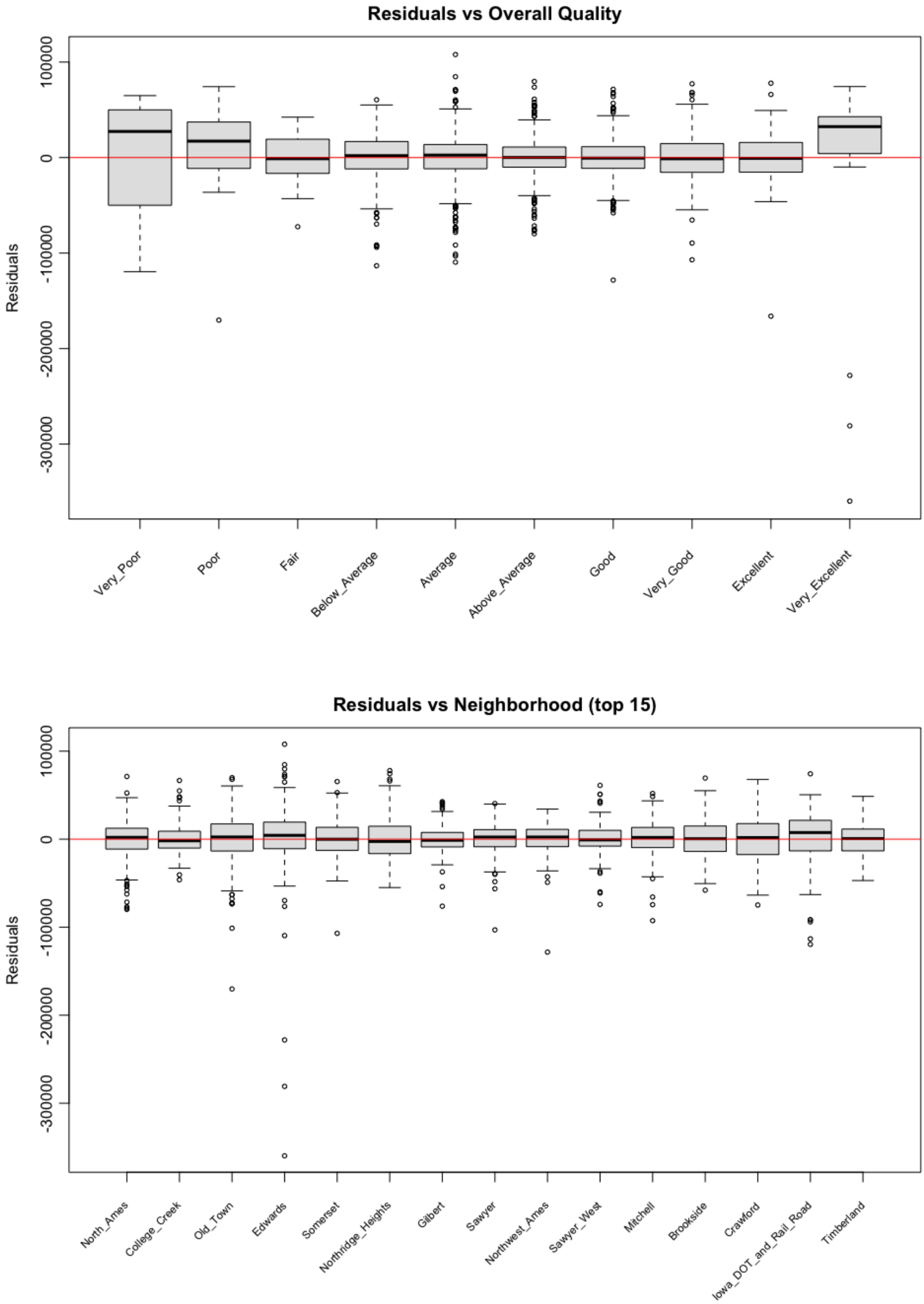


Figure 17: Residuals Versus All Categorical Predictors of Model 3 (And Take Top 15 Neighborhoods in the plot of Residuals Versus Neighborhood).

Table 6: Variance Inflation Factors for Numerical Predictors in Model 3.

Predictor	**VIF**
Gr_Liv_Area	4.402
TotRms_AbvGrd	3.885
Full_Bath	2.099
Bedroom_AbvGr	1.991
Age_ames	1.630
Total_Bsmt_SF	1.525
Fireplaces	1.384
Garage_Area_log1p	1.196
Lot_Area	1.150

Model 4: Box-Cox Transformation$_{\lambda=0.222}$(Sale Price$_i$)= β_0 + β_1Gross Living Area$_i$+ β_2Total Basement Area$_i$ + β_3Log(1 + Garage area$_i$) + β_4Age of House$_i$ + β_5Overall Quality$_i$ + β6Neighborhood$_i$ + β_7Lot Area$_i$ + β_8Full Bath$_i$ + β_9Bedroom Above Ground$_i$ + β_{10}Fireplaces$_i$ + β_{11}Total Rooms Above Ground$_i$ + ε_i

Based on Model 3, problematic observations were identified using standardized residuals, leverage, and Cook's distance. High-leverage observations that were neither outliers nor influential were retained; observations were removed only if they were outliers, bad leverage points, or influential observations. This resulted in the removal of 165 unique observations (Tables 7 and 8). As shown in Table 7, all removals were driven by Cook's distance greater than 4/n: all 13 outliers and all 6 bad leverage points were also included among the 165 influential observations. Table 8 shows that 2,765 observations remained after removal.

Model diagnostics improved significantly after deleting outlier observations. The response fitted plot (Figure 18) is tighter now; residual (Figure 19) plots are more randomly spread about zero, and the normal Q-Q plot has less divergence from the reference line and shows less extension into the tails. The histogram of the residuals is less heavily-tailed. The boxplots of the two categorical predictors (Figure 20) have smaller differences in their vertical spread across the groups, therefore, we will be using the reduced dataset for subsequent model-selection steps.

Table 7: Influence-Diagnostic Thresholds and Screening Results for Model 3.

Diagnostic	**Cutoff**	**Count**
High leverage (h_ii > 2(p + 1)/n)	0.0314	253
Outlier (|std resid| > 4)	4.0000	13
Bad leverage (high lev & outlier)	0.0314	6
Influential (Cook's D > 4/n)	0.0014	165
Removed (union of bad leverage + influential)	NA	165

Table 8: Sample Size Before and After Removing Flagged Observations.

Quantity	**Count**
Removed observations	165
Remaining observations	2765

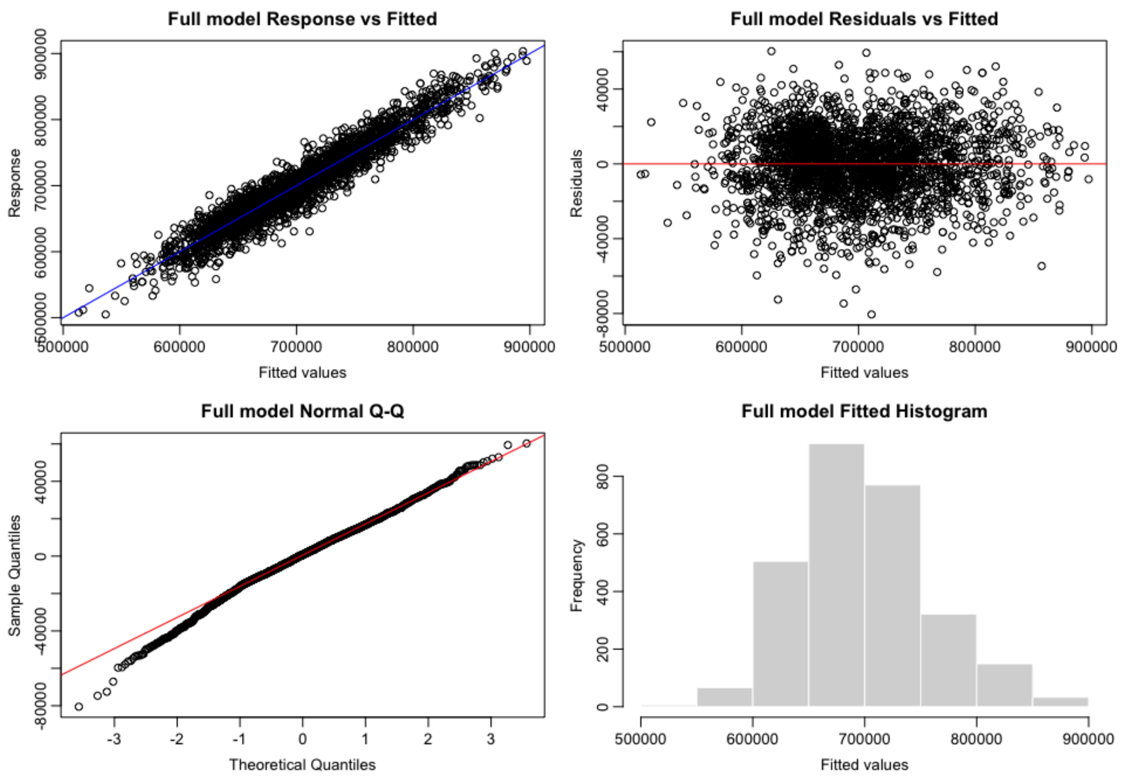


Figure 18: Response Versus Fitted Values, Residuals Plot, Fitted Histogram, Normal Q-Q Plot of Model 4.

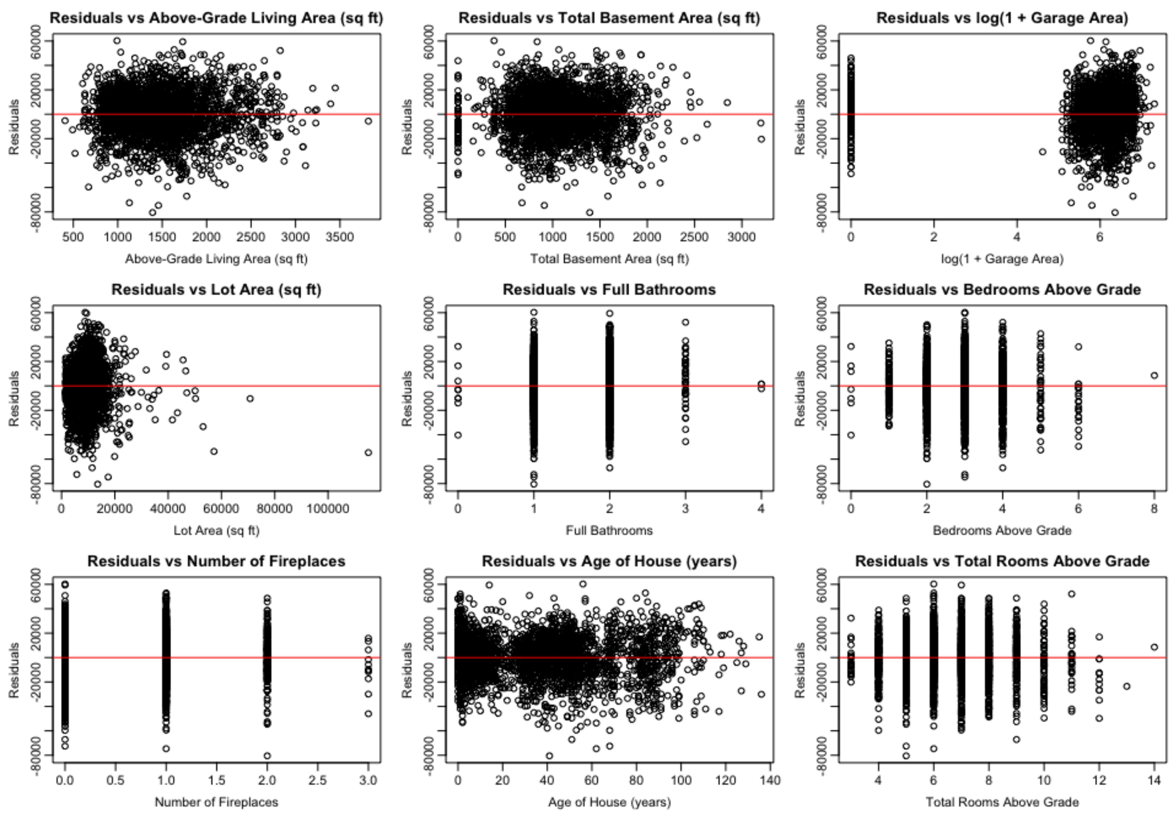


Figure 19: Residuals Versus All Numerical Predictors of Model 4.

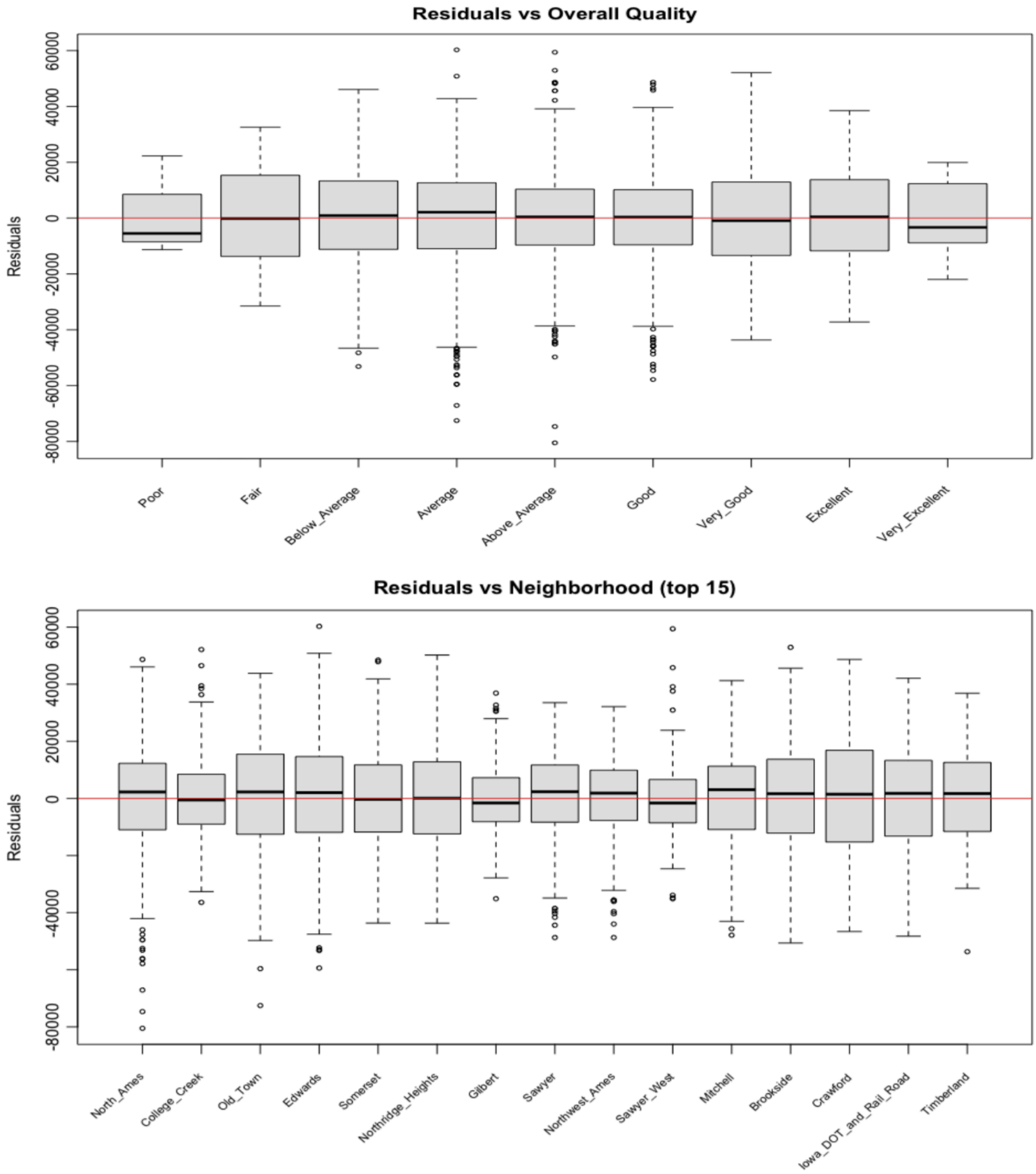


Figure 20: Residuals Versus All Categorical Predictors of Model 4 (And Take Top 15 Neighborhoods in the plot of Residuals Versus Neighborhood).

Model 5: Box-Cox Transformation$_{\lambda=0.222}$(Sale Price$_i$)= β_0 + β_1Gross Living Area$_i$ + β_2Total Basement Area$_i$ + β_3Log(1 + Garage area$_i$) + β_4Age of House$_i$ + β_5Overall Quality$_i$ + β_6Neighborhood$_i$ + β_7Lot Area$_i$ + β_8Bedroom Above Ground$_i$ + β_9Fireplaces$_i$ + β_{10}Total Rooms Above Ground$_i$ + ε_i

Model 6: Box-Cox Transformation$_{\lambda=0.222}$(Sale Price$_i$)= β_0 + β_1Gross Living Area$_i$ + β_2Total Basement Area$_i$ + β_3Log(1 + Garage area$_i$) + β_4Age of House$_i$ + β_5Overall Quality$_i$ + β_6Neighborhood$_i$ + β_7Lot Area$_i$ + β_8Bedroom Above Ground$_i$ + β_9Fireplaces$_i$ + ε_i

Backward elimination was performed using AIC as the primary criterion to select the best model. Full Bath was removed in Model 5 first, then following removing Total Rooms Above Ground from Model 6. As illustrated in Table 9, removing Full Bath resulted in a reduction in the AIC from 54229.41 to 54227.42 and the BIC from 54490.11 to 54482.19. Removing Total Rooms Above Ground next resulted in another reduction to the AIC of 54225.79 and BIC of 54474.63. During all model building steps, the adjusted R-squared value remained constant at 0.9155 and the information criteria decreased. Therefore, the removal of Full Bath and Total Rooms Above Ground did not add substantial additional explanation of variance. Also, the diagnostic plots for models 5 and 6 (Figures 21-26) remained consistent because Total Rooms Above Ground had the second highest VIF value (Table 6). Because Model 6 was simpler and had the lowest AIC and BIC value, but had the same adjusted R-squared value, Model 6 was considered to be the final model.

A final VIF check (Table 10) also showed that all VIF values were below 5, with the largest VIF only 2.201 for Gross Living Area, indicating that serious multicollinearity was not a concern.

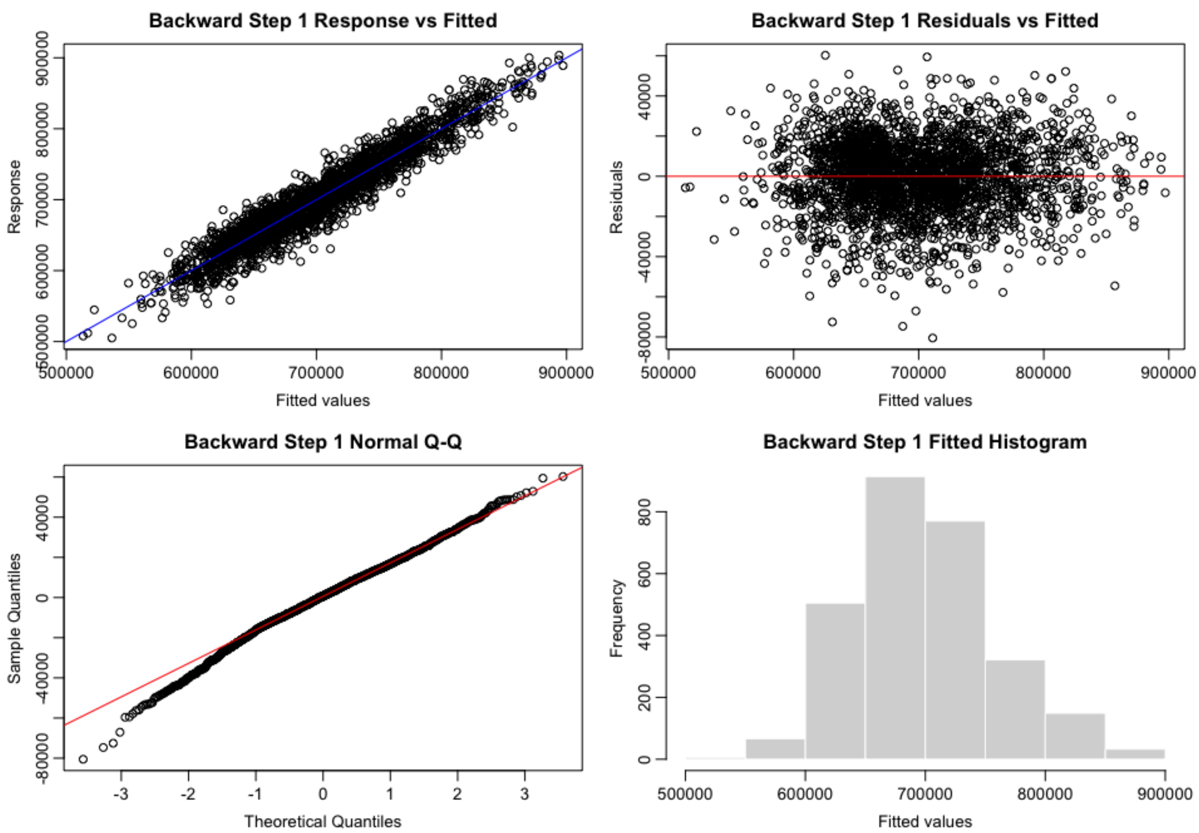


Figure 21: Response Versus Fitted Values, Residuals Plot, Fitted Histogram, Normal Q-Q Plot of Model 5.

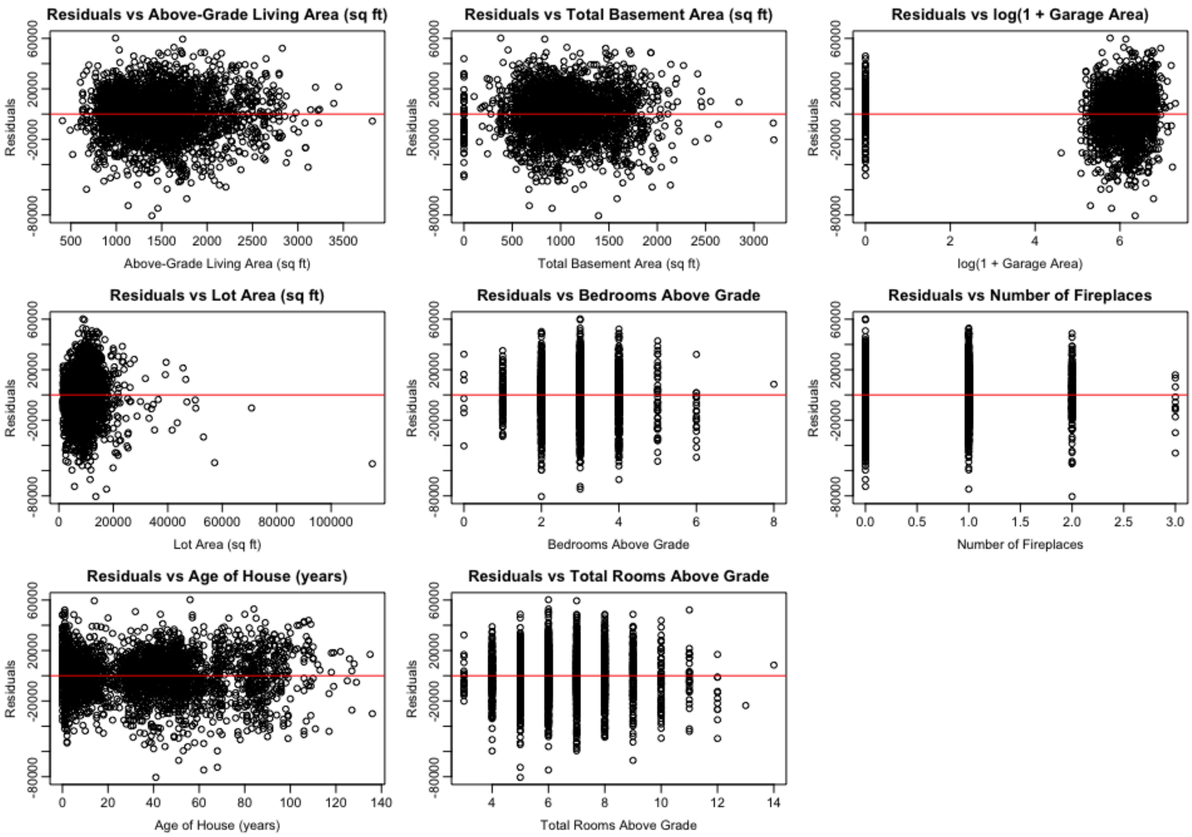
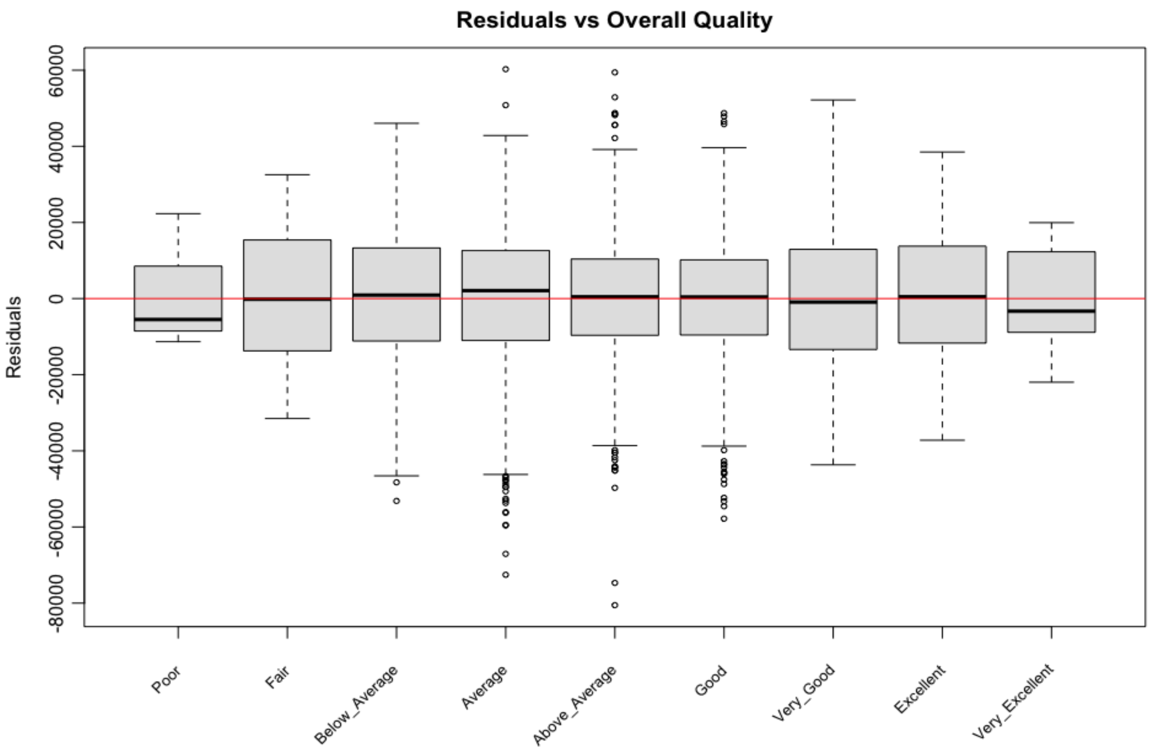


Figure 22: Residuals Versus All Numerical Predictors of Model 5.



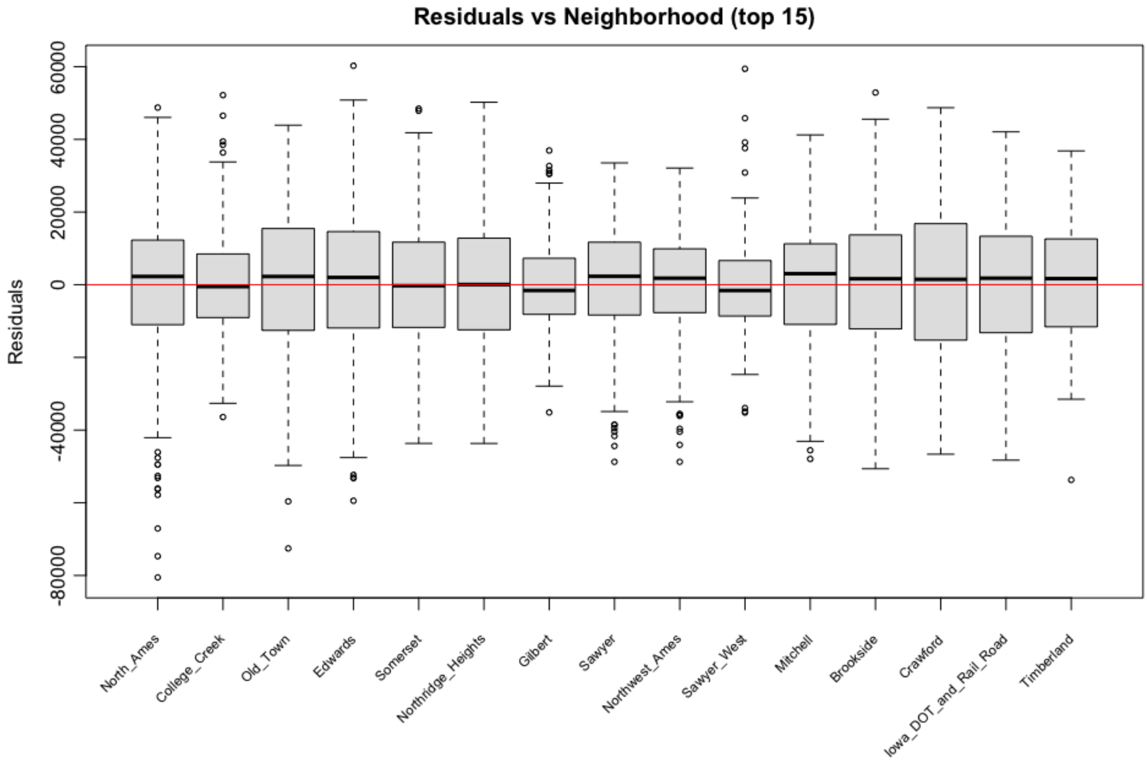


Figure 23: Residuals Versus All Categorical Predictors of Model 5 (And Take Top 15 Neighborhoods in the plot of Residuals Versus Neighborhood).

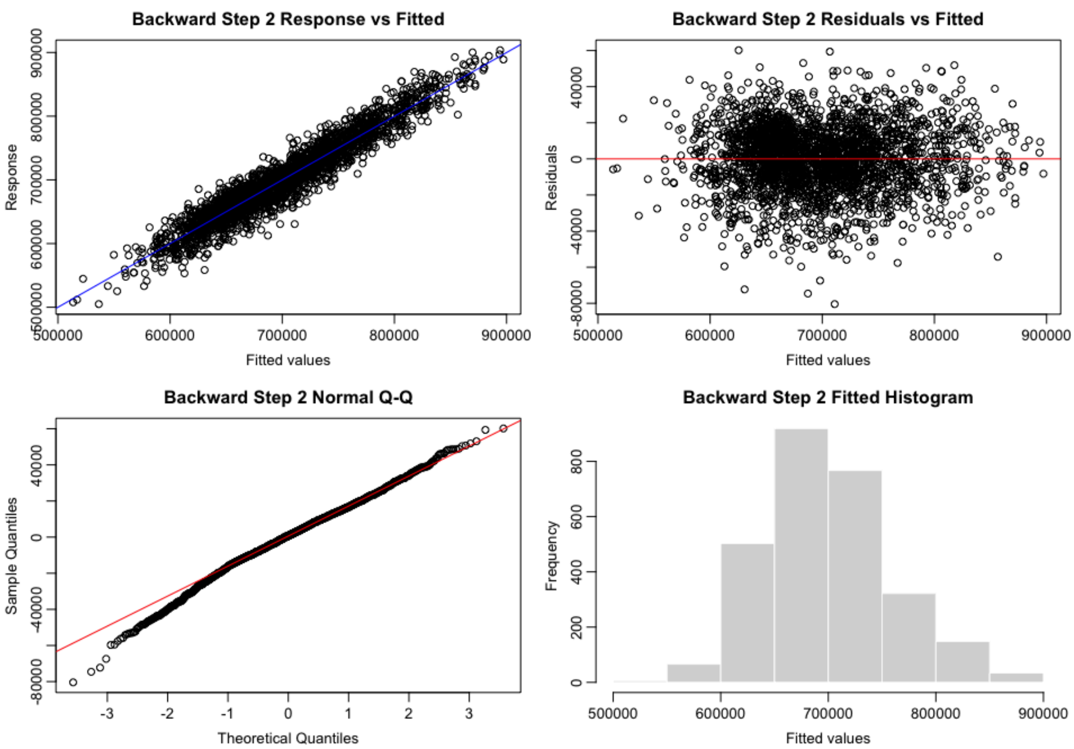


Figure 24: Response Versus Fitted Values, Residuals Plot, Fitted Histogram, Normal Q-Q Plot of Model 6.

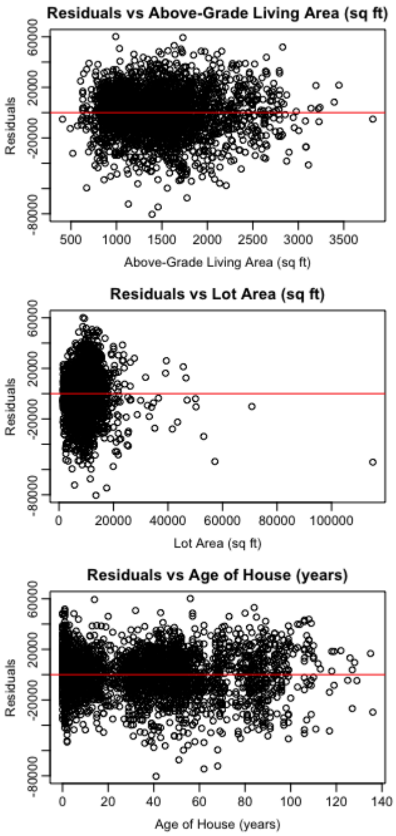
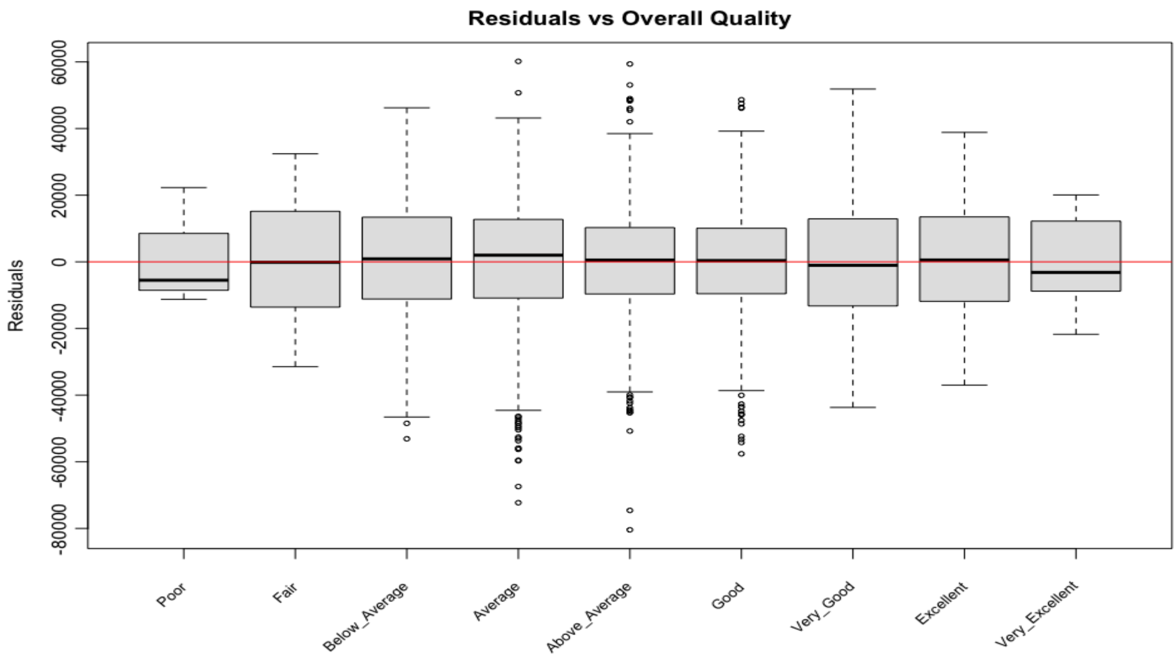


Figure 25: Residuals Versus All Numerical Predictors of Model 6.



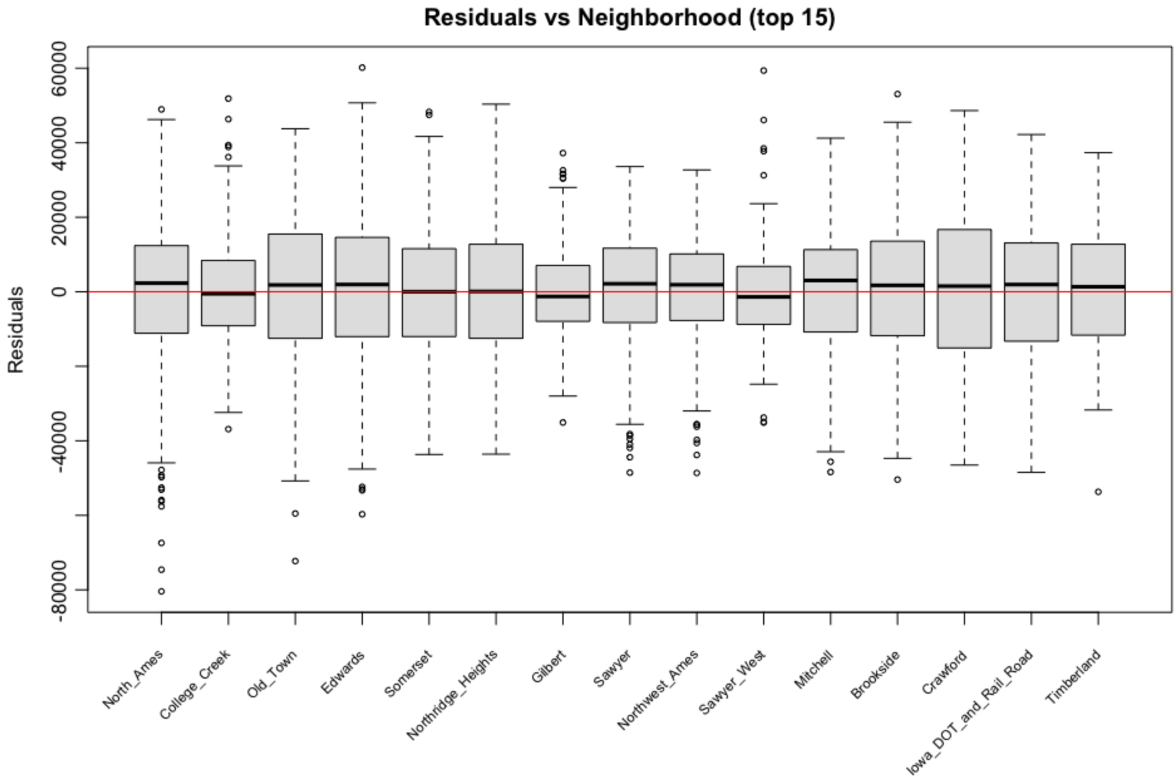


Figure 26: Residuals Versus All Categorical Predictors of Model 6 (And Take Top 15 Neighborhoods in the plot of Residuals Versus Neighborhood).

Table 9: Comparison of Candidate Models During AIC-Based Backward Selection on the Cleaned Data.

Model	**Degree of Freedom**	**Adj_R2**	**AIC**	**AICc**	**BIC**
Full model	43	0.9155	54229.41	54230.94	54490.11
Model 5	42	0.9155	54227.42	54228.88	54482.19
Model 6	41	0.9155	54225.79	54227.18	54474.63

Table 10: Variance Inflation Factors for Numerical Predictors in the Final Selected Model.

Predictor	**VIF**
Gr_Liv_Area	2.201

Bedroom_AbvGr	1.594
Age_ames	1.328
Total_Bsmt_SF	1.397
Fireplaces	1.364
Garage_Area_log1p	1.190
Lot_Area	1.179

Cross Validation

Following the selection of model 6, we utilized an 80/20 holdout validation and a ten-fold cross-validation to evaluate this model. According to Table 11, these results indicated that the RMSE for the final two models would be less than that of the initial model on both the test set and ten-fold CV. For the test, the RMSE for the original transformed scale reduced from 18555.55 to 18063.95 and ten-fold CV RMSE reduced from 19003.47 to 18148.23. Therefore, the final model proved to have greater validation performance with a more simplistic form.

Table 11: Holdout-Test and 10-Fold Cross-Validation RMSE for the Cleaned-Data Baseline and Final Selected Models

Model	Test_n_used	Test_RMSE_transformed	CV10_RMSE_transformed	Test_RMSE_price
Initial model	553	18555.55	19003.47	20305.84
Final model	553	18063.95	18148.23	20221.02

Final Model Versus Initial Model

The final model (Figure 27-29) had obvious improvements over the original model. The response versus fitted scatter plot improved its fit to referential by being close to the diagonal line, and there were less points off the primary pattern for the higher fitted values, the residuals plots for both of the fitted variables, total basement area, the above ground living area, and house age all show a more random distribution around zero. For the Garage Area, there was still a residual cluster for zero area garages. However, after removing an influential observation and applying log transformation, residuals were less impacted by extreme residuals. For the two categorical predictors, the residual boxplots became more balanced, with smaller differences in vertical spread across groups. The Normal Q-Q plot was much closer to the reference line, with fewer large deviations in both tails. The residual histogram also became more symmetric and closer to a normal shape. Numerically, the adjusted R-squared increased from 0.9071 to 0.9155, while AIC decreased from 54486.88 to 54225.79 and BIC decreased from 54717.95 to 54474.63 (Table 12).

Table 12: Comparison of the Cleaned-Data Baseline Model and Final Selected Model on the Same Box–Cox Response Scale.

Model	Predictors	Adj_R2	AIC	AICc	BIC
Initial model	38	0.9071	54486.88	54488.08	54717.95
Final model	41	0.9155	54225.79	54227.18	54474.63

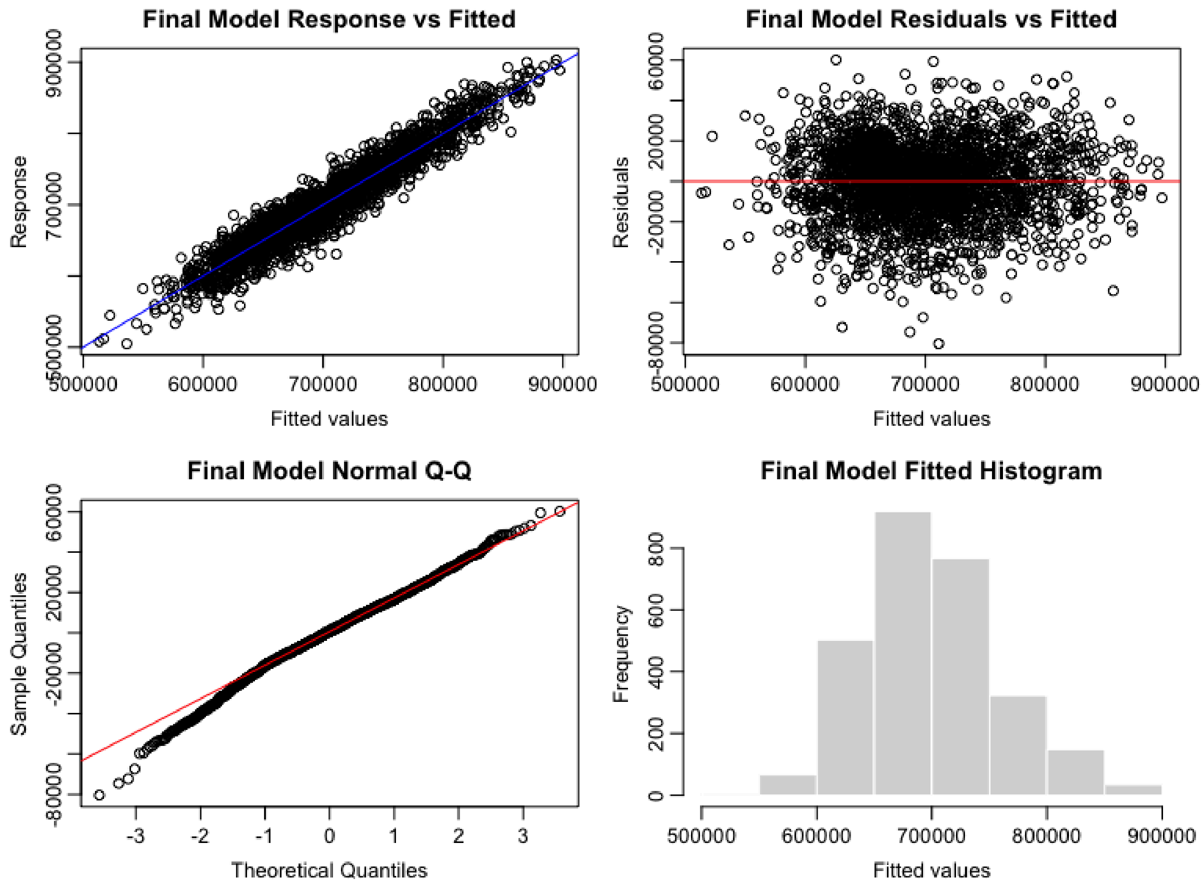


Figure 27: Response Versus Fitted Values, Residuals Plot, Fitted Histogram, Normal Q-Q Plot of Final Model.

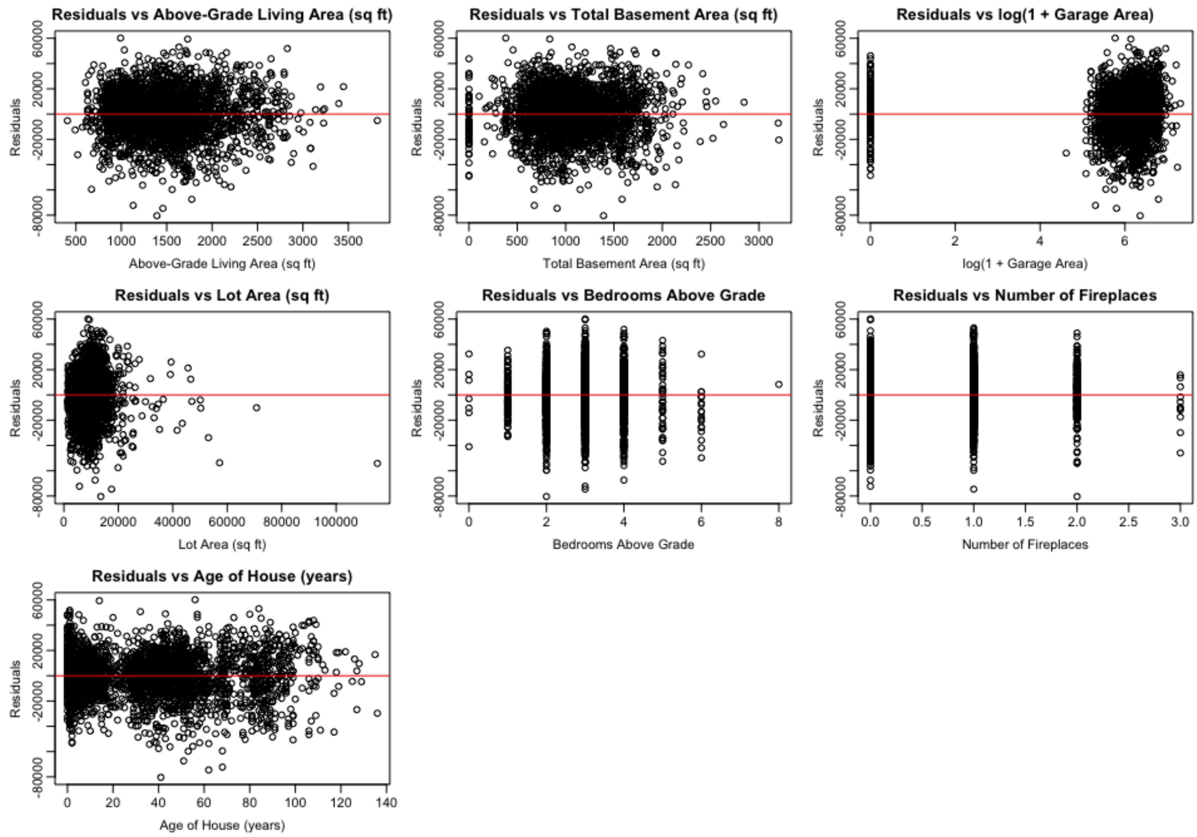
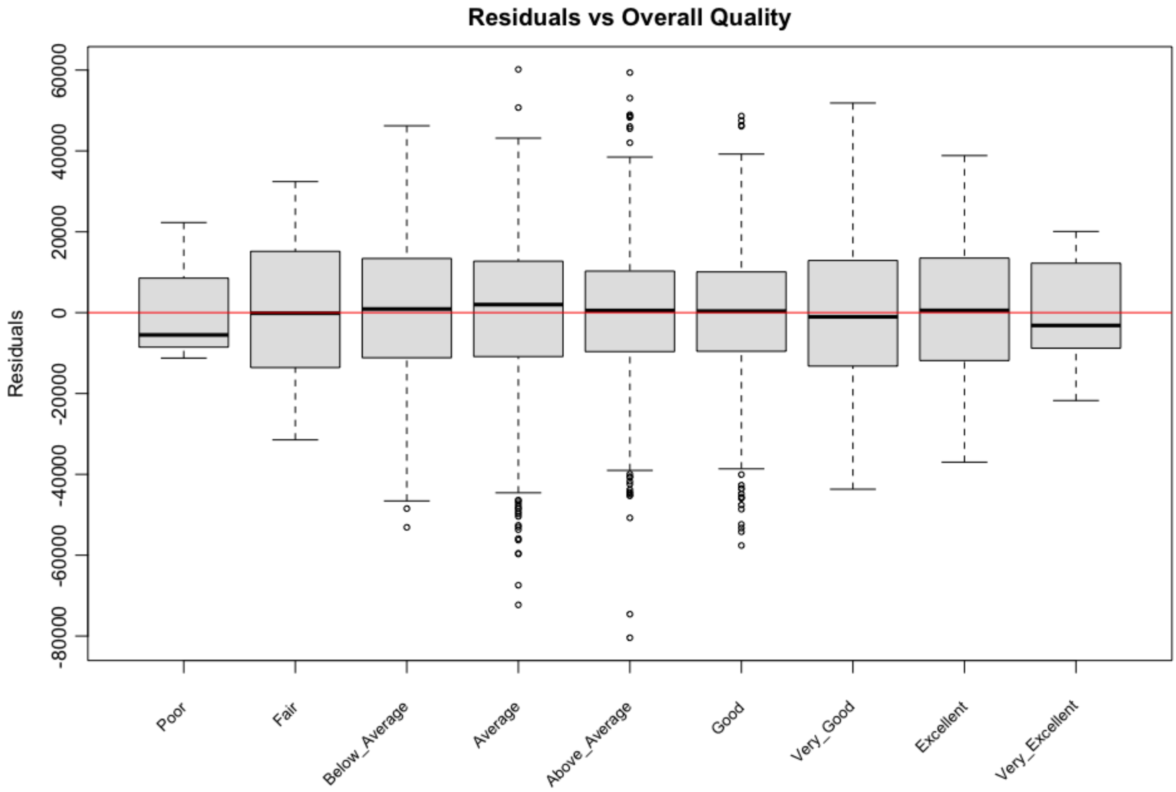


Figure 28: Residuals Versus All Numerical Predictors of Final Model.



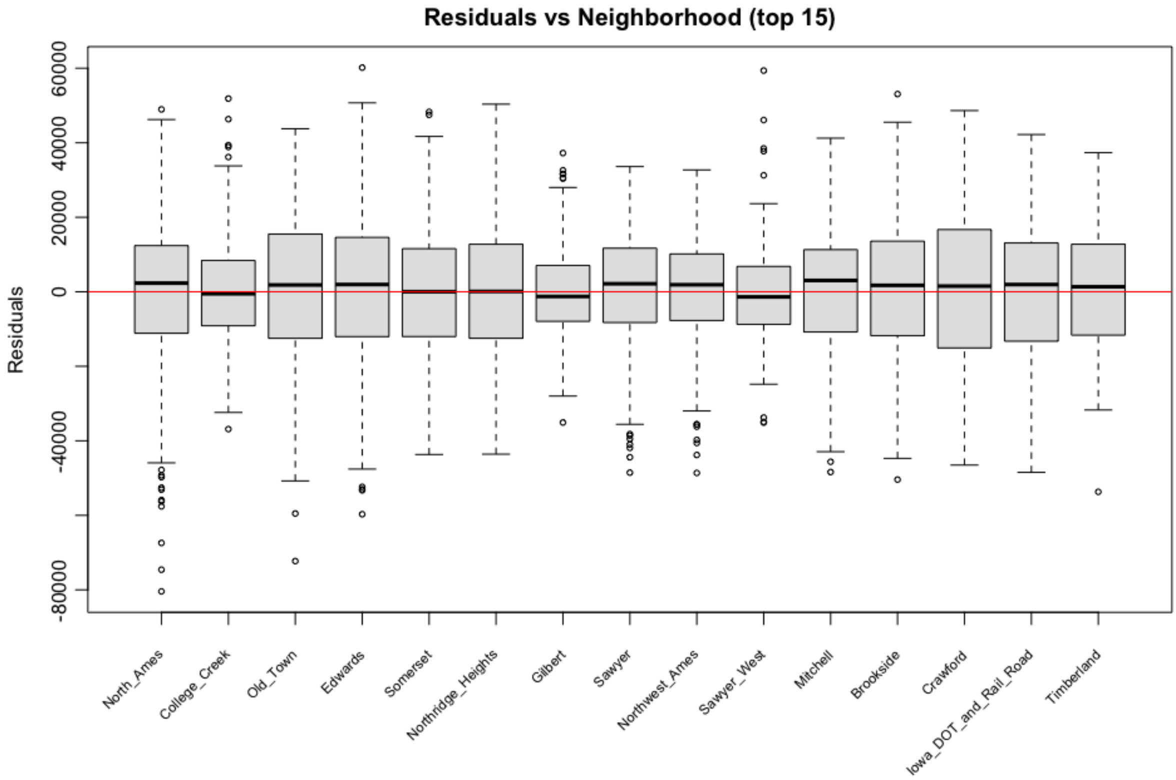


Figure 29: Residuals Versus All Categorical Predictors of Final Model (And Take Top 15 Neighborhoods in the plot of Residuals Versus Neighborhood).

Final Model Inference and Results

Table 13: Final Model Summary (Coefficient Estimates, Standard Errors, 95% Confidence Intervals, and p-Values for the Final Box–Cox Transformed Sale Price Model).

(R-squared = 0.9168, Adjusted R-squared = 0.9155, F-statistic: 731.8 on 41 and 2723 DF, p-value < 0.001)

Term	Estimate	Std_Error	CI_Lower	CI_Upper	p_value
(Intercept)	501377.64	9317.95	483106.67	519648.62	< 0.001
Gr_Liv_Area	45.48	1.32	42.90	48.06	< 0.001
Total_Bsmt_SF	20.63	1.12	18.43	22.83	< 0.001
Garage_Area_log1p	3001.12	286.51	2439.32	3562.92	< 0.001
Lot_Area	1.00	0.08	0.84	1.16	< 0.001
Bedroom_AbvGr	-3188.93	591.13	-4348.04	-2029.82	< 0.001
Fireplaces	6629.38	668.11	5319.33	7939.43	< 0.001
Age_ames	-448.97	26.80	-501.53	-396.42	< 0.001
Overall_QualFair	48837.48	9623.53	29967.31	67707.64	< 0.001
Overall_QualBelow_Average	73316.57	9173.69	55328.48	91304.67	< 0.001

Overall_QualAverage	87059.87	9125.27	69166.72	104953.01	< 0.001
Overall_QualAbove_Average	96577.80	9143.19	78649.50	114506.10	< 0.001
Overall_QualGood	107201.74	9185.83	89189.83	125213.65	< 0.001
Overall_QualVery_Good	126651.46	9282.88	108449.27	144853.66	< 0.001
Overall_QualExcellent	158464.52	9505.38	139826.03	177103.01	< 0.001
Overall_QualVery_Excellent	166471.42	10838.95	145218.03	187724.80	< 0.001
NeighborhoodCollege_Creek	6632.09	1756.77	3187.36	10076.82	< 0.001
NeighborhoodOld_Town	-4623.82	1835.69	-8223.31	-1024.34	0.01183
NeighborhoodEdwards	-7387.35	1673.74	-10669.28	-4105.43	< 0.001
NeighborhoodSomerset	11821.44	2088.93	7725.40	15917.49	< 0.001
NeighborhoodNorthridge_Heights	17647.12	2351.35	13036.51	22257.73	< 0.001
NeighborhoodGilbert	1734.65	2029.54	-2244.94	5714.24	0.39280
NeighborhoodSawyer	210.83	1730.11	-3181.64	3603.29	0.90300
NeighborhoodNorthwest_Ames	1266.55	1910.38	-2479.39	5012.48	0.50740
NeighborhoodSawyer_West	893.34	2075.58	-3176.52	4963.20	0.66690
NeighborhoodMitchell	2588.73	2028.22	-1388.28	6565.74	0.20190
NeighborhoodBrookside	2389.67	2147.59	-1821.40	6600.74	0.26590
NeighborhoodCrawford	23481.66	2157.75	19250.66	27712.65	< 0.001
NeighborhoodIowa_DOT_and_Rail_Road	-9177.53	2440.94	-13963.80	-4391.25	< 0.001
NeighborhoodTimberland	11034.22	2651.45	5835.17	16233.28	< 0.001
NeighborhoodNorthridge	19649.61	2825.55	14109.18	25190.05	< 0.001
NeighborhoodStone_Brook	19106.37	3565.91	12114.21	26098.52	< 0.001
NeighborhoodSouth_and_West_of_Iowa_State_University	-6394.97	3031.08	-12338.42	-451.52	0.03497
NeighborhoodClear_Creek	5710.97	3299.91	-759.62	12181.56	0.08363
NeighborhoodMeadow_Village	-23579.55	3353.83	-30155.85	-17003.26	< 0.001
NeighborhoodBriardale	-28575.04	3521.65	-35480.43	-21669.66	< 0.001
NeighborhoodBloomington_Heights	-2582.54	3932.17	-10292.87	5127.79	0.51140
NeighborhoodVeenker	12557.38	4722.85	3296.65	21818.10	0.00789
NeighborhoodNorthpark_Villa	-9056.25	4023.11	-16944.90	-1167.60	0.02446
NeighborhoodBlueste	-17511.18	6170.72	-29610.94	-5411.42	0.00458
NeighborhoodGreens	4042.77	7113.36	-9905.36	17990.91	0.56990
NeighborhoodLandmark	-12995.96	18068.29	-48424.90	22432.98	0.47200

Model Summary

Table 13 summarizes the final fitted model. The response variable Sale Price used Box-Cox-Transformed, and predictor garage area used logarithm transformation which defined as log(1 + Garage_Area) to correct for skewness and heteroscedasticity.

The final model includes 9 predictor terms, Gross Living Area, Total Basement Area, Log-Transformed Garage Area, Number of Bedroom Above Ground, Fireplaces, Age, Lot Area, Overall Quality and Neighborhood, resulting in 41 estimated coefficients including the intercept.

Interpretation of Coefficients

- Intercept: the intercept of 501377.64 represents the expected Box-Cox transformed sale price when all numerical predictors are equal to zero, and the categorical predictors are at their baseline levels. However, in the practical context of the housing market, this value represents an extreme extrapolation beyond the sample data range, as a residential home cannot realistically have zero square footage, zero rooms, or zero age. Therefore, the intercept does not possess a meaningful real-world economic interpretation.

- Gross Living Area: Each additional square foot of gross living area increases the transformed price by 45.48 units on average with all other predictors held fixed. To ease the uncertainty because of the sampling method, we are 95% confident that the transformed sale price increases by between 42.90 to 48.06 units for each additional square foot of living area when all other predictors are held fixed.

- Total Basement Area: Each additional square foot of basement area increases the expected transformed price by 20.63 units with all other predictors held fixed. To ease the uncertainty because of the sampling method, we are 95% confident that the transformed sale price increases by between 18.43 to 22.83 units for each additional square foot of basement area when all other predictors are held fixed.

- Log-transformed Garage Area: Since Garage Area is log-transformed as log(1 + Garage_Area), a one-unit increase in the log-transformed predictor does not correspond to a constant additive change in the original scale. Instead, the predictor scale changes multiplicatively, multiplying the “1 + Garage Area” by *e* is associated with an expected increase of approximately 3001.12 units on average in the transformed sale price, holding all other predictors fixed, and has 95% confident the increased transformation Sales Price interval is between 2439.32 to 3562.92 units.

- Lot Area: We expect the transformed sale price to increase by 1.00 units on average for additional square feet of lot area with all other predictors held fixed. To ease the uncertainty because of the sampling method, we are 95% confident that the transformed sale price increases by between 0.84 to 1.16 units for each additional square feet of lot area when all other predictors are held fixed.

- Number of Bedroom Above Ground: We expect the transformed sale price to decrease by 3188.93 units on average when each additional bedroom above ground with all other predictors held fixed. To ease the uncertainty because of the sampling method, we are 95% confident that the transformed sale price decreases by between 2029.82 to 4348.04 units for each additional bedroom above ground when all other predictors are held fixed.

- Fireplaces: Each additional fireplace increases the transformed price by 6629.38 units on average with all other predictors held fixed. To ease the uncertainty because of the sampling method, we are 95% confident that the transformed sale price increases by between 5319.33 to 7939.43 units for each additional fireplace when all other predictors are held fixed.

- Age: Each additional year on the age of the home decreases the transformed price by 448.97 units on average with all other predictors held fixed. To ease the uncertainty because of the sampling method, we are 95% confident that the transformed sale price decreases by between 396.42 to 501.53 units for a one year older home when all other predictors are held fixed.

- Overall Quality: Relative to a "Poor" home and holding all other predictors fixed, the expected Box–Cox transformed sale price is higher by 48837 ("Fair"), 73317 ("Below Average"), 87060 ("Average"), 96578 ("Above Average"), 107202 ("Good"), 126651 ("Very Good"), 158465 ("Excellent"), and 166471 ("Very Excellent") units on average. Each level is significant at $p < 0.001$, and the estimates rise monotonically with quality, so quality has a strong, ordered, independent association with price.

- Neighborhood: Moving from North Ames to College Creek, the expected increase in the transformed price is 6632.09 units on average, holding all other predictors fixed. Moving to a different neighborhood may have significant negative effects, as shown in Table 13 for Northpark Village (-9056.25) and Briardale (-28575.04). While only select neighborhoods are highlighted here for brevity, the remaining neighborhood coefficients are interpreted analogously: each quantifies the expected change in the transformed sale price compared to the reference neighborhood of North Ames, assuming all other predictors remain fixed.

All continuous predictors show the expected direction and statistical significance (p-values < 0.001). The gross living area, basement area, lot area, fireplace, and log-transformed garage area are positively associated with the transformed sale price, while the age of home and numbers of bedrooms above ground decrease it. Moreover, overall quality remains a powerful predictor which is consistent with our hypothesis.

For neighborhoods, while several areas such as the College Creek show large positive correlation with the transformed sale price, the negative coefficients observed for some neighborhoods such as Briardale should be interpreted with caution. These estimates are partly driven by very small sample sizes. For instance, Briardale contains only thirty observations and Green Hills has just two, which leads to unstable estimates and wide confidence intervals. Nonetheless, location clearly influences housing prices, although the precision of individual neighborhood effects varies considerably.

Comparison with Literature

Consistent with Yang (2025), we identify Overall_Qual, Gr_Liv_Area, and garage-related features as key positive drivers of housing prices. While Yang employed machine learning models for predictive accuracy, our focus on interpretability via linear regression shows the same directional findings.

Our results also echo Aziz et al. (2021). Their hedonic pricing model found that structural, locational, and neighborhood variables significantly influence property values, with larger structures and better locations commanding higher prices. Our neighborhood coefficients strongly support their conclusion that there are both positive and negative effects in their model, and "negative results due to preference about neighborhood services and varying property characteristics"(Aziz et al., 2021) also explains the negative estimated coefficients of Neighborhood predictors.

Model Performance Assessment

The final model explains 91.68% of the variation in the Box–Cox transformed sale price. Because R^2, AIC, and BIC are only comparable across models fitted to the same response on the same data. So we compare to a baseline that uses the original six predictors refitted on the same cleaned data and the same Box–Cox response (Table 11). In that fair comparison, adjusted R^2 rises from 0.9071 to 0.9155, and AIC and BIC fall from 54,486.88 to 54,225.79 and from 54,717.95 to 54,474.63, respectively. The drop in AIC and BIC shows the improvement survives a penalty for the three additional predictors, so the gain reflects better fit rather than added complexity. Relative to the proposal's untransformed model, the improvement is in assumption satisfaction: the transformations and influential-point removal resolved the non-constant variance and non-normal tails seen in the preliminary model.

Discussion and conclusion

Main Conclusions

This analysis set out to describe how structural and neighbourhood characteristics of residential properties in Ames, Iowa are associated with sale price, and to test the hypothesis that overall construction quality remains a statistically significant predictor of price after controlling for the other characteristics of a home. Our final Multiple linear regression model answers both questions clearly.

After applying a Box–Cox transformation to sale price, a log transformation to garage area, removing influential observations, and reducing the predictor set through AIC-based backward elimination, the model explains approximately 91.7% of the variation in the transformed sale price (adjusted $R^2 = 0.9155$).

Within this model, every level of overall quality is highly significant ($p < 0.001$) even after living area, basement area, garage area, lot area, bedrooms, fireplaces, age, and neighbourhood are held fixed. This directly supports the idea that construction quality is not merely a proxy for size or location, but also carries a large and independent association with price.

Overall, the final model provides a strong and interpretable framework for understanding housing prices in Ames, Iowa and highlights the importance of construction quality, property size, and location in determining residential value.

Key Findings

The strongest and most consistent finding is the effect of overall quality. Holding all other predictors fixed, the expected transformed price rises monotonically across quality grades, with the gap between the lowest and highest grades exceeding 160,000 transformed units (Table 13). Because every quality level is significant at the 0.1% level, quality behaves as a genuine, independent driver of price rather than an artifact of correlated size variables — the partial effect a buyer is paying for is construction quality itself.

The size and structural predictors behave as expected and are all significant ($p < 0.001$). Gross living area, total basement area, lot area, the number of fireplaces, and (log-transformed) garage area are each positively associated with price. In contrast, the age of the home is negatively associated with it (about −449 transformed units per year). One result that merits comment is bedrooms above grade, whose coefficient is negative (−3188.93) once gross living area is in the model. This is not contradictory: holding total living area fixed, dividing that area into more bedrooms implies smaller, less valuable rooms, so the partial effect of an additional bedroom is negative even though larger homes (which tend to have more bedrooms) cost more. This illustrates the value of a multiple-regression interpretation, where each coefficient is a conditional effect.

Neighbourhood is also influential, but its effect is heterogeneous. Relative to the baseline (North Ames), several neighbourhoods command large, significant premiums (for example Northridge Heights and Crawford). In contrast, others are associated with significant discounts (for example Briardale and Meadow Village). However, several neighbourhood estimates are statistically insignificant and several have very wide confidence intervals, reflecting small sample sizes in those areas. Taken together, the results confirm that location matters, but they also caution that individual neighbourhood effects are estimated with very different levels of precision.

These findings are broadly consistent with the literature. Like Yang (2025), We find that overall quality, living area, and garage-related features are positively associated with sale price. Our analysis further shows that these relationships remain statistically significant after controlling for other housing characteristics in a multiple linear regression model. Our neighbourhood results echo the hedonic framework of Aziz et al. (2021), in which structural and locational attributes jointly shape property values and location can contribute either positively or negatively depending on local characteristics.

Limitations

Several limitations should be considered when interpreting these results. First, this analysis identifies statistical associations rather than causal relationships. Because the data are observational, we cannot conclude that changing a housing characteristic would directly affect sale price. In addition, the dataset includes only homes sold in Ames, Iowa, between 2006 and 2010. Therefore, the results may not generalize to other housing markets or more recent time periods.

Second, although the Box–Cox and log transformations improved the model diagnostics, some assumption violations remain. The residual-versus-fitted plot became more uniform, and the Normal Q–Q plot moved closer to the reference line. However, slight departures are still visible, especially in the tails of the Q–Q plot. As a result, the standard errors, confidence intervals, and p-values should be viewed as approximations rather than exact measures.

We also removed 165 influential observations to improve model stability. This helped produce a better-fitting model for typical homes in the dataset. However, the final model may perform less well for unusually priced or atypical properties. In addition, different criteria for identifying influential observations can yield slightly different coefficient estimates and conclusions.

Finally, multicollinearity was only assessed for the numerical predictors, and all VIF values were below 5. Potential collinearity involving categorical predictors cannot be completely ruled out. Some potentially important variables were also unavailable, including school quality, household income, mortgage rates, and proximity to employment centres. The omission of these variables may have introduced omitted-variable bias and reduced the model's explanatory power.

Future Improvements

Future work could incorporate additional predictors of neighbourhood amenities, accessibility, and local economic conditions, thereby reducing omitted-variable bias and providing a fuller picture of price formation.

Extending the analysis to multiple cities and more recent transactions would test whether the relationships observed in Ames hold more broadly.

Methodologically, allowing interactions (for example between quality and living area) or comparing the linear model against more flexible approaches could reveal nonlinearities that an additive linear specification cannot capture. Finally, the precision of the neighbourhood effects could be improved by grouping very small neighbourhoods into broader, more stable categories, tightening the corresponding confidence intervals while preserving the locational signal the analysis has shown to be important.

Recommendations and Practical Implications

For the research question — how structural and neighbourhood characteristics relate to sale price — the practical message is that quality and usable size are the dominant value levers, while location acts as a substantial but neighbourhood-specific modifier. For homeowners and sellers, investments that raise a property's overall quality grade are associated with the largest expected gains in value, more so than adding bedrooms without adding a living area. For developers, the results suggest that investments in construction quality may provide larger returns than simply increasing the number of bedrooms. This finding is consistent with the strong and significant effect of overall quality observed in the final model. For buyers and appraisers, the model offers an interpretable baseline for what a home "should" cost given its measurable characteristics, and the large neighbourhood coefficients quantify the location premium or discount attached to a given area.

These implications should be read as descriptive associations that hold on average for the Ames market, and the imprecise neighbourhood estimates (those with wide intervals or small samples) should be used only as rough guides rather than precise adjustments.

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